

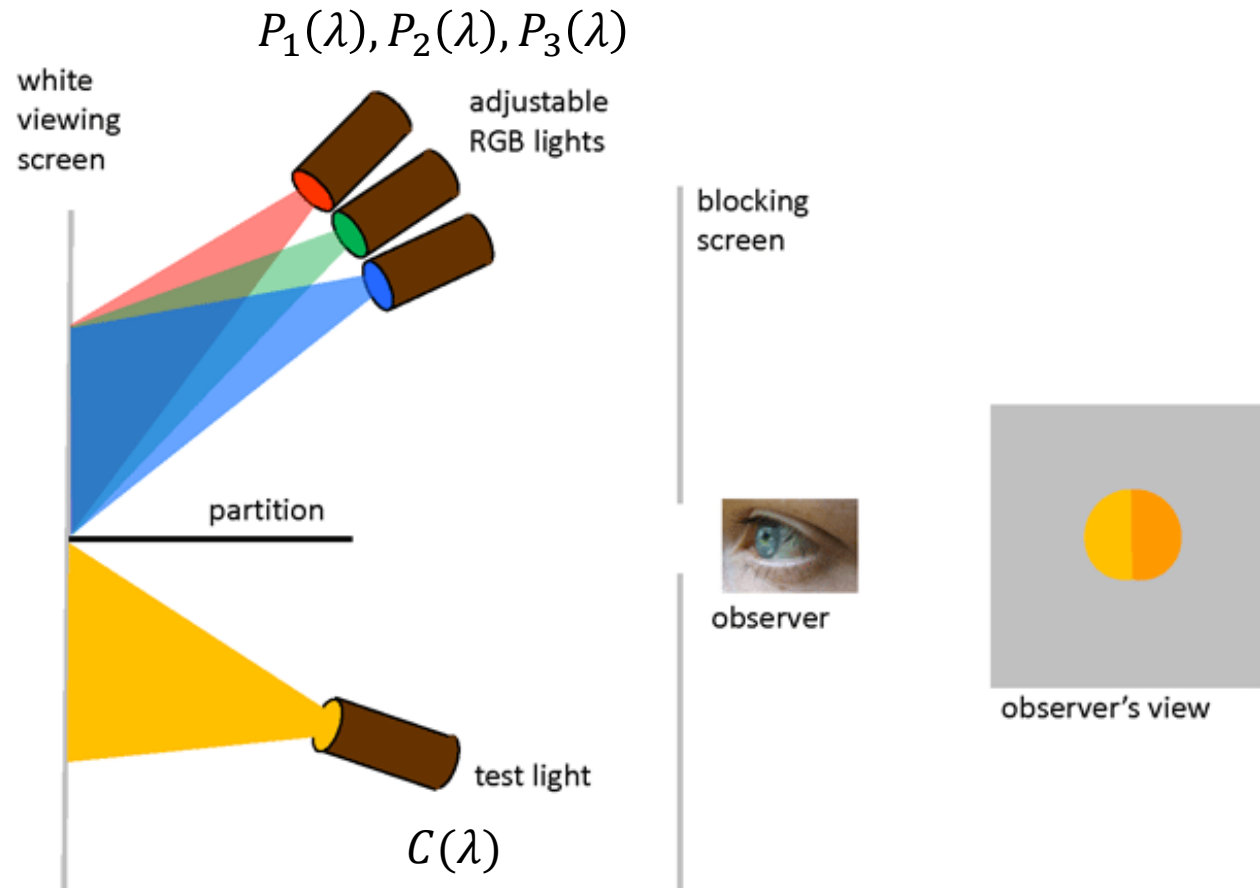
ECE 253

Guest lecture #2 on Color

Pamela Cosman

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Color Matching Experiment



- **Test light:** $C(\lambda)$ has any spectral distribution. Observer cannot change it.
- **Primary lights:** $P_1(\lambda)$, $P_2(\lambda)$, $P_3(\lambda)$ have any* spectral distributions. Observer can adjust the overall gain for each one separately, but cannot change the shape of the distribution
- **Goal:** Adjust the intensity factors until 2 sides look identical
- **Results:** For any* test light, can get a match! Radically different spectra on the 2 sides, look the same: *Metamers*
- The match is *unique*

It's a Linear System!

Satisfies homogeneity:

$$\text{If } T_1P_1 + T_2P_2 + T_3P_3 \leftrightarrow C(\lambda)$$

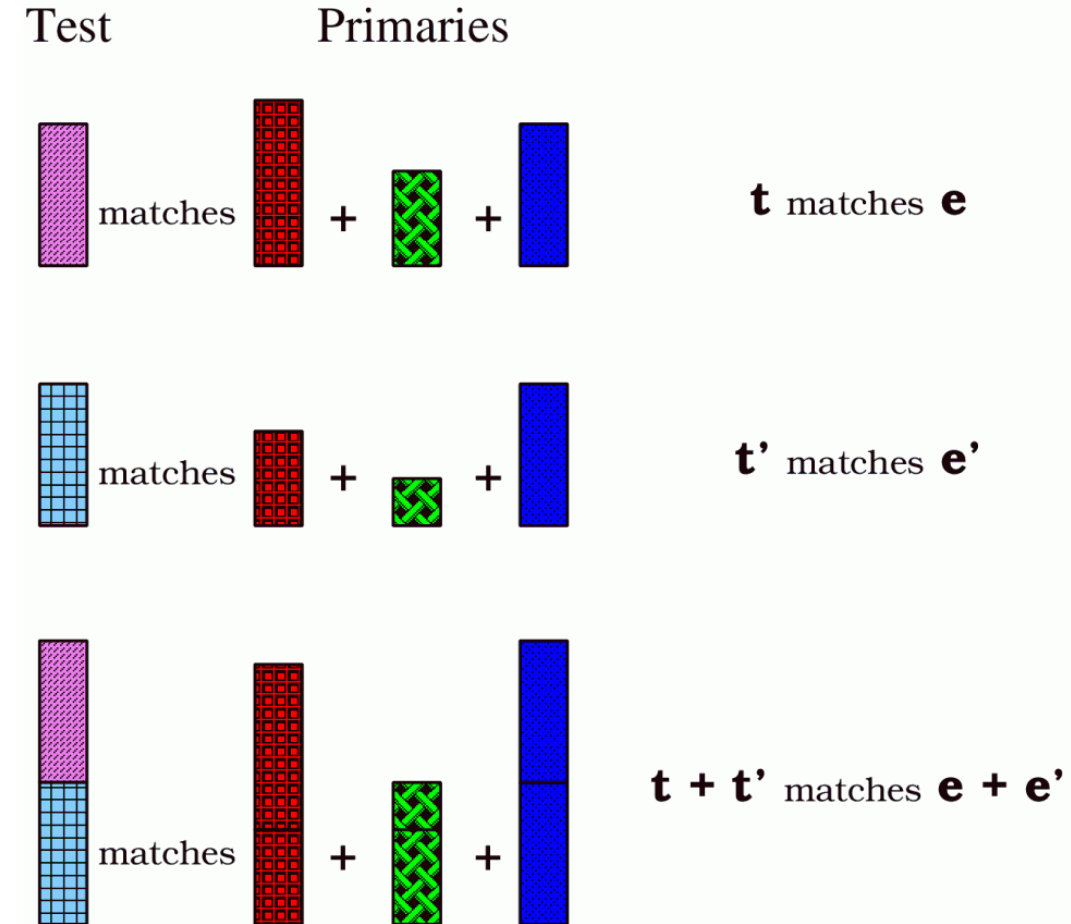
$$\text{then } aT_1P_1 + aT_2P_2 + aT_3P_3 \leftrightarrow aC(\lambda)$$

And superposition:

$$\text{If } T_1P_1 + T_2P_2 + T_3P_3 \leftrightarrow C(\lambda)$$

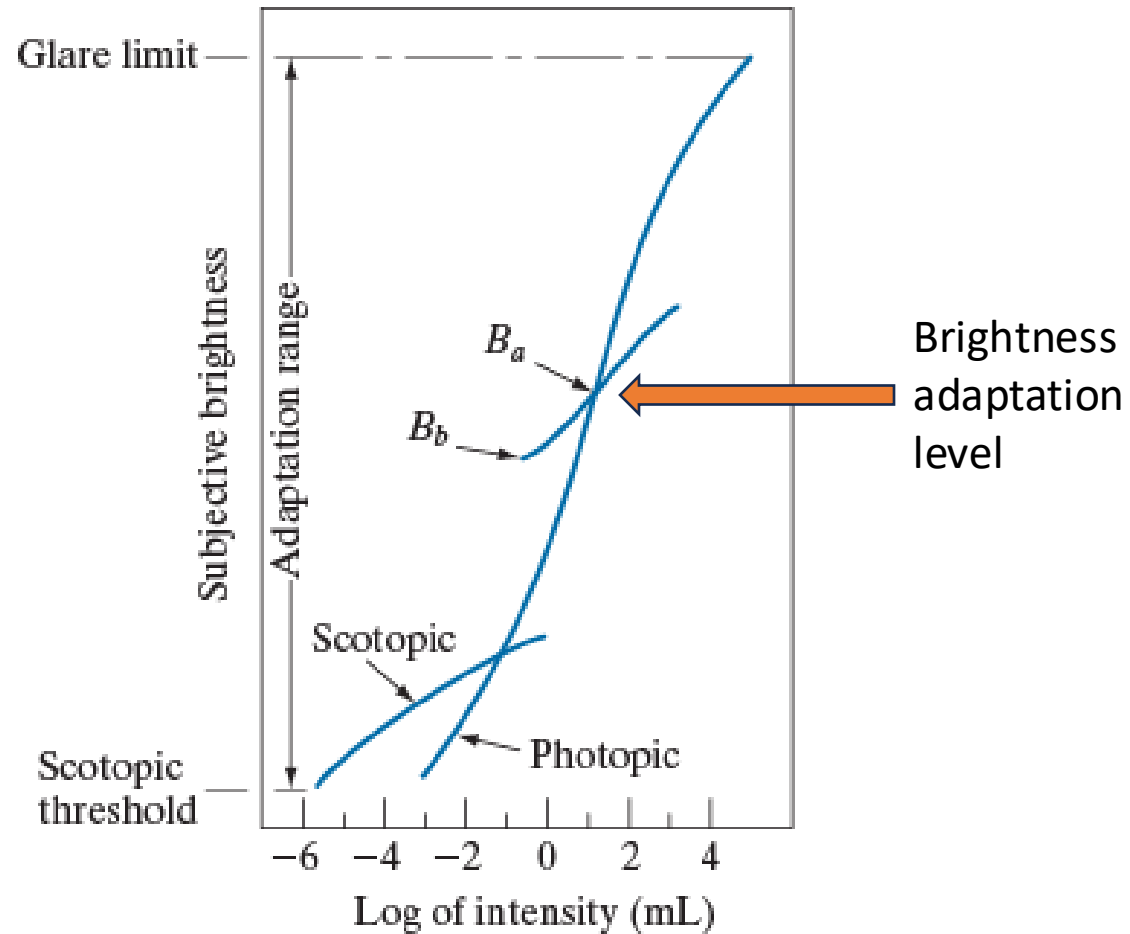
$$\text{and } T'_1P_1 + T'_2P_2 + T'_3P_3 \leftrightarrow C'(\lambda)$$

$$\text{then } (T_1 + T'_1)P_1 + (T_2 + T'_2)P_2 + (T_3 + T'_3)P_3 \leftrightarrow C(\lambda) + C'(\lambda)$$

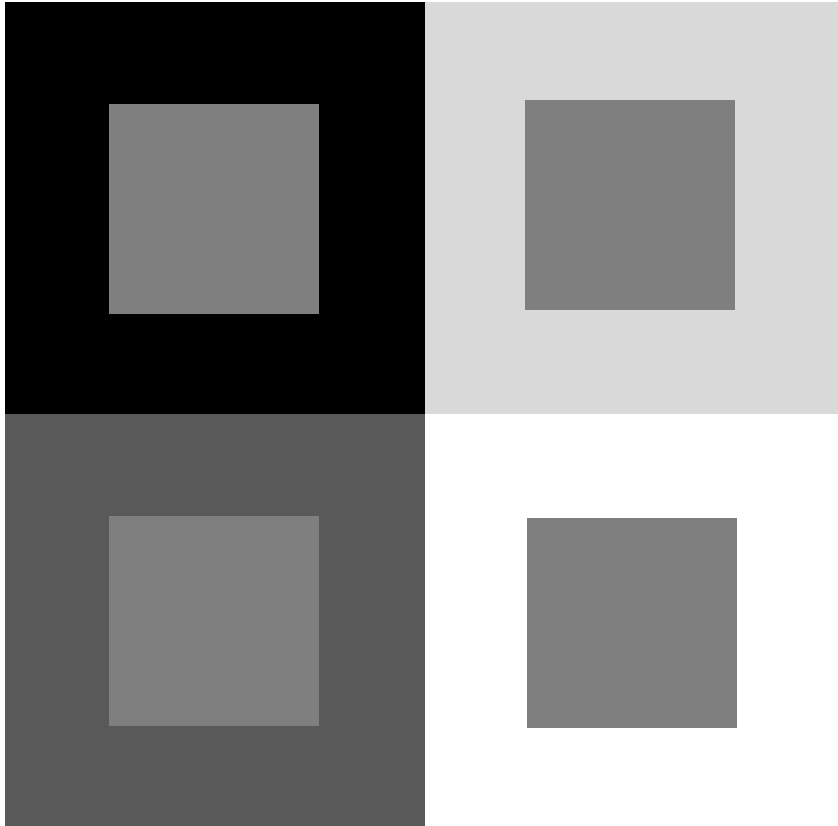


Rods and Cones – Light Adaptation

- Scotopic vision uses rods, dim light vision
- Photopic vision uses cones, bright light vision

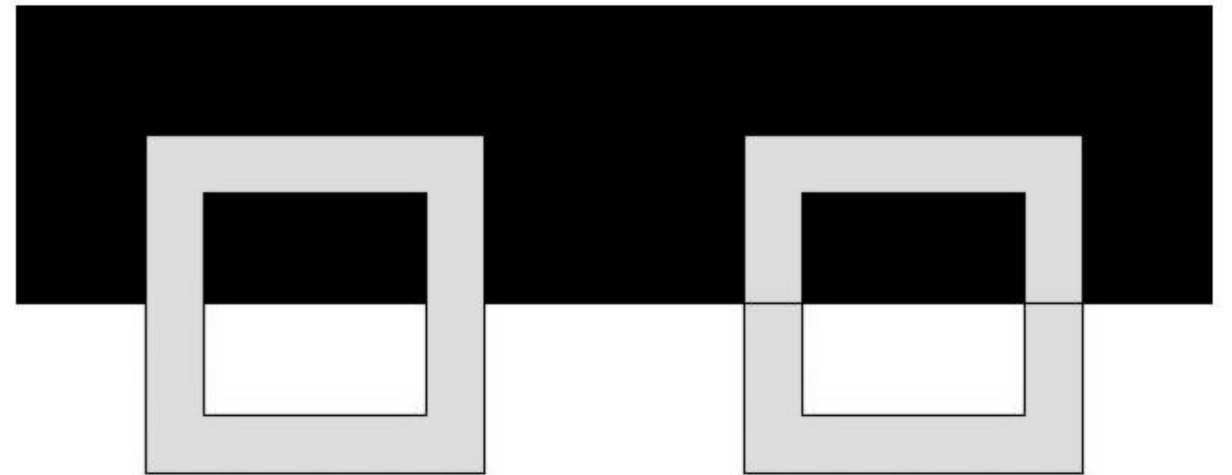


Brightness perception depends on context

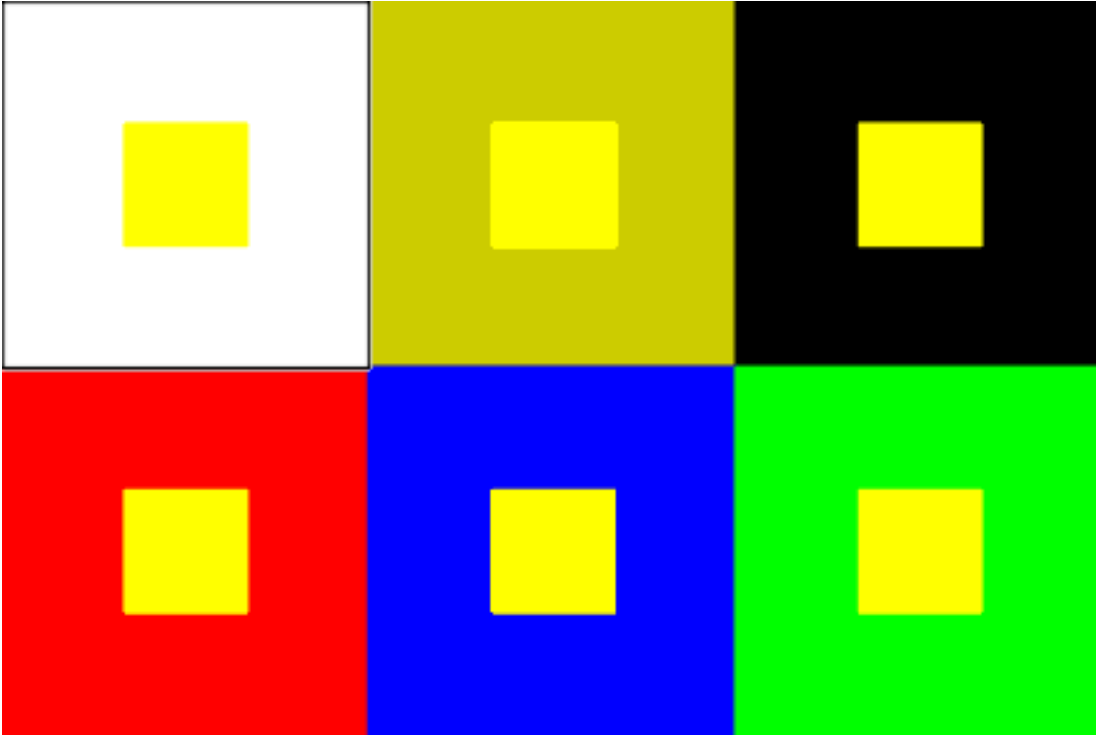


Simultaneous contrast

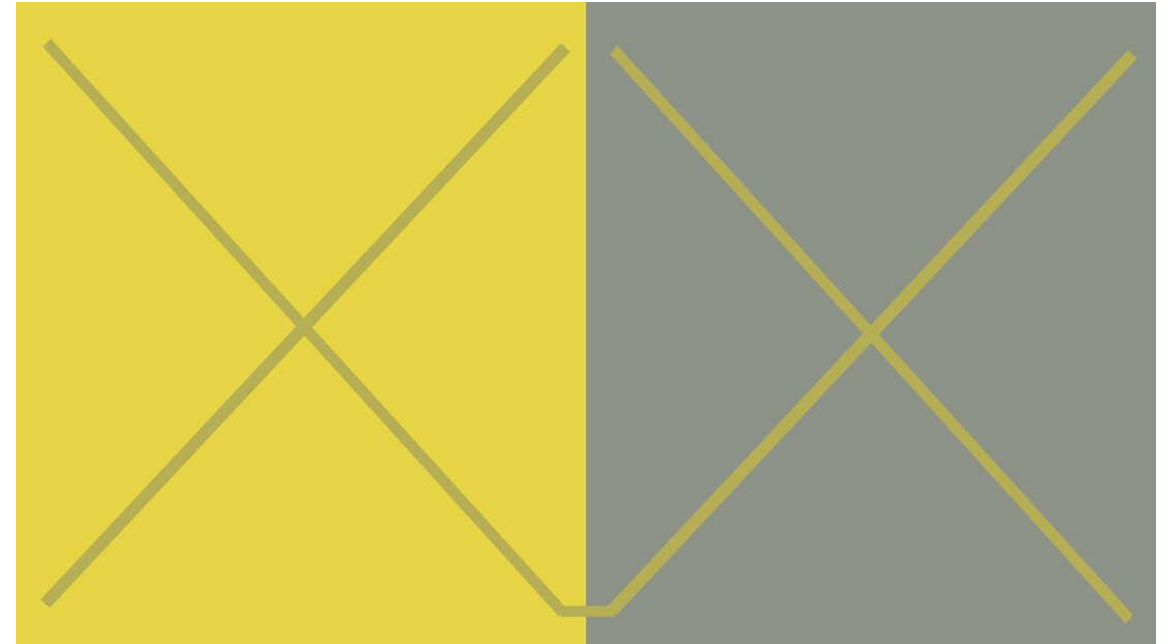
Benussi ring



Color perception also depends on context



<https://www.colorcube.com/illusions/scindctn.htm>



Source: Josef Albers Interaction of Color

Assumption sneaks into the color matching experiment: The spectrum of the light arriving at your eye completely determines the perceived color

In reality, it depends on context and adaptation

Connection between behavior and physiology

matrix of primaries

$$P = \begin{bmatrix} P_{1,1} & P_{2,1} & P_{3,1} \\ P_{1,2} & P_{2,2} & P_{3,2} \\ \vdots & \vdots & \vdots \\ P_{1,n} & P_{2,n} & P_{3,n} \end{bmatrix}$$

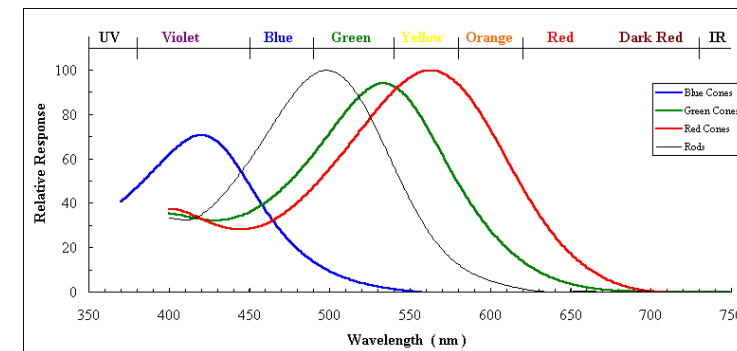
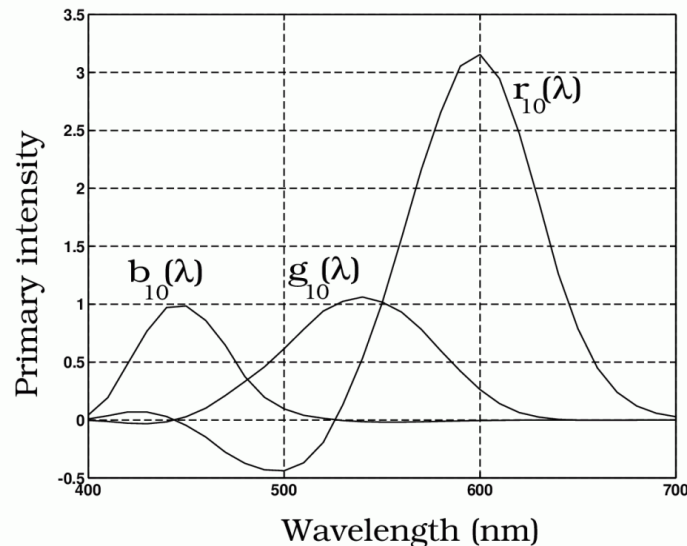
$$M = (RP)^{-1}R$$

$$\begin{bmatrix} m_{1,1} & \dots & m_{1,i} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,i} & \dots & m_{2,n} \\ m_{3,1} & \dots & m_{3,i} & \dots & m_{3,n} \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} R_{1,1} & \dots & R_{1,i} & \dots & R_{1,n} \\ R_{2,1} & \dots & R_{2,i} & \dots & R_{2,n} \\ R_{3,1} & \dots & R_{3,i} & \dots & R_{3,n} \end{bmatrix}$$

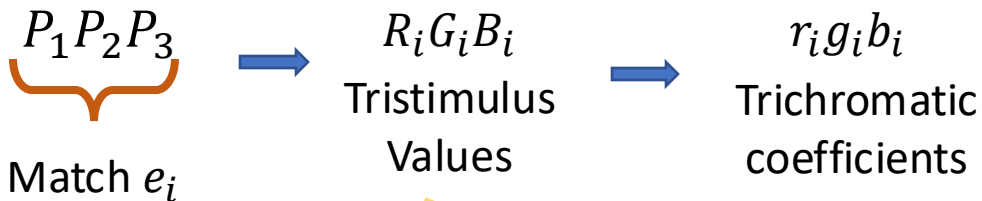
Color Matching Functions

3x3
matrix

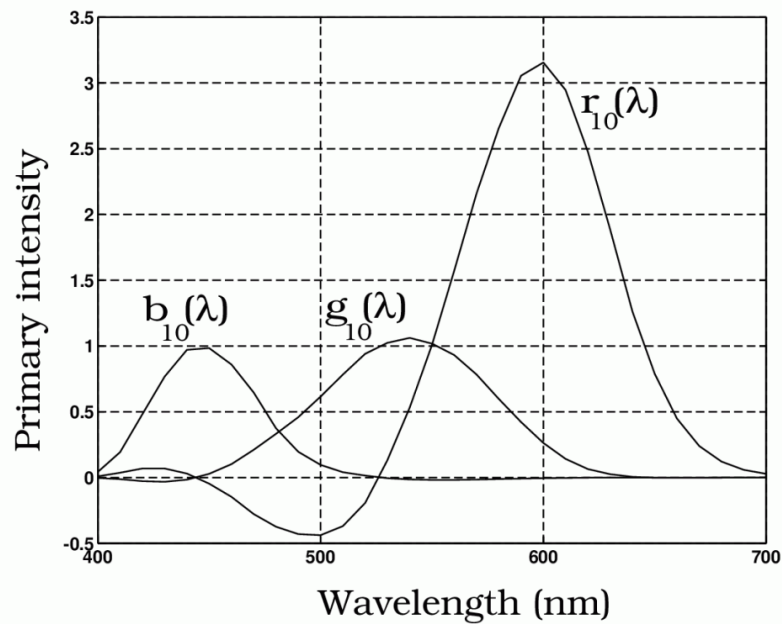
Spectral absorption
functions of the cones



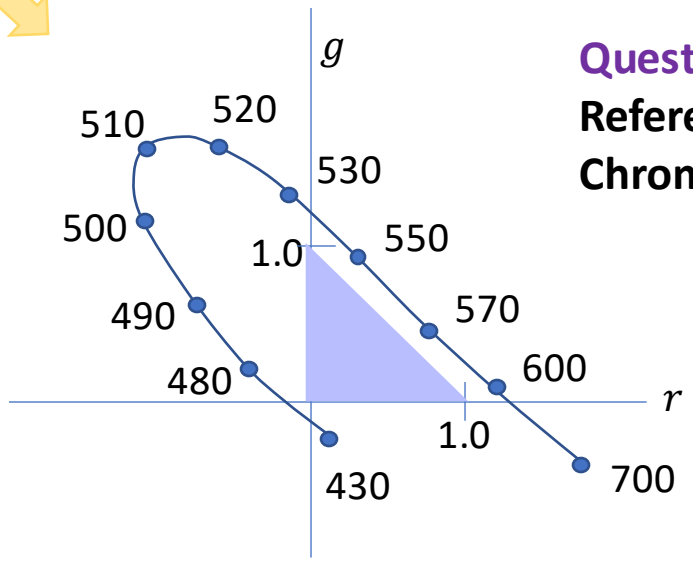
Chromaticity Diagram from coordinates r,g



$$r_i = \frac{R_i}{R_i + G_i + B_i} \quad g_i = \frac{G_i}{R_i + G_i + B_i}$$



Color Matching Functions



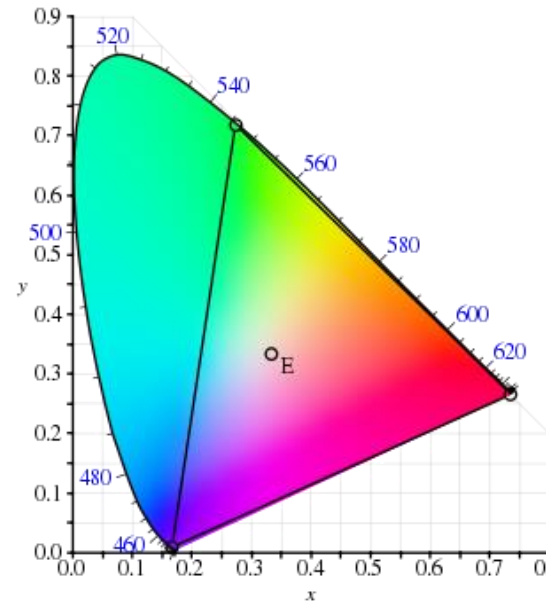
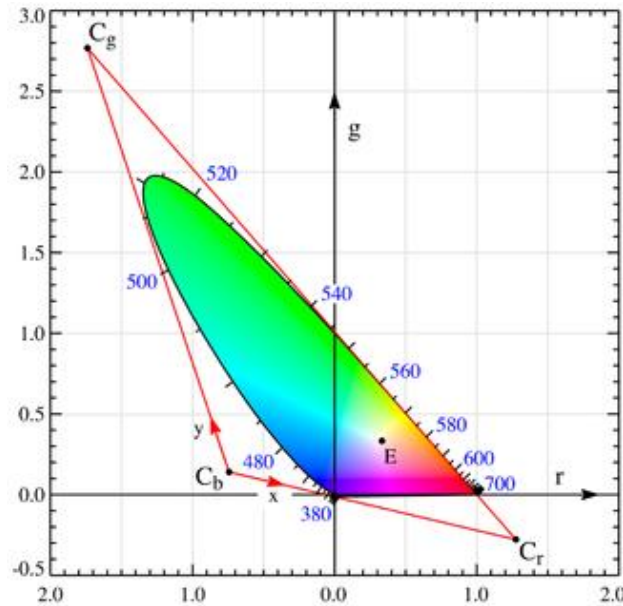
Question: Where is the Reference White in this Chromaticity Diagram?

Chromaticity Diagram

CIE XYZ Color Coordinate System

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.49000 & 0.32000 & 0.2 \\ 0.17697 & 0.81240 & 0.01063 \\ 0 & 0.01 & 0.99000 \end{bmatrix} \begin{bmatrix} R_c \\ G_c \\ B_c \end{bmatrix}$$

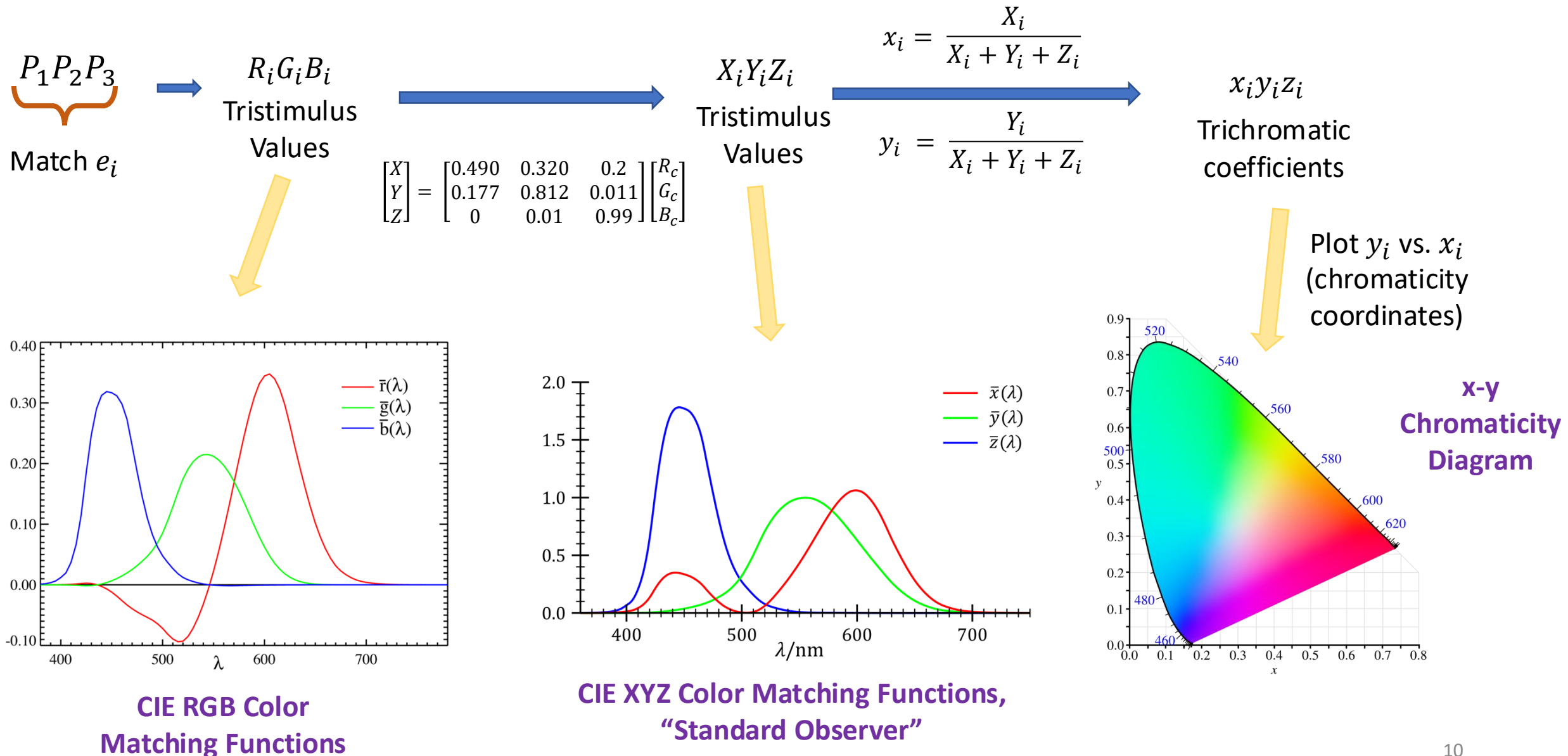
- Mathematical exercise that rotates the chromaticity plot so that everything is in the positive quadrant
- X, Y, Z are always positive
- x, y, z are always positive
- Known as the CIE XYZ system
- Developed in 1931, still the main coordinate system used today for referencing other color systems
- The primaries that correspond to this system are physically unrealizable



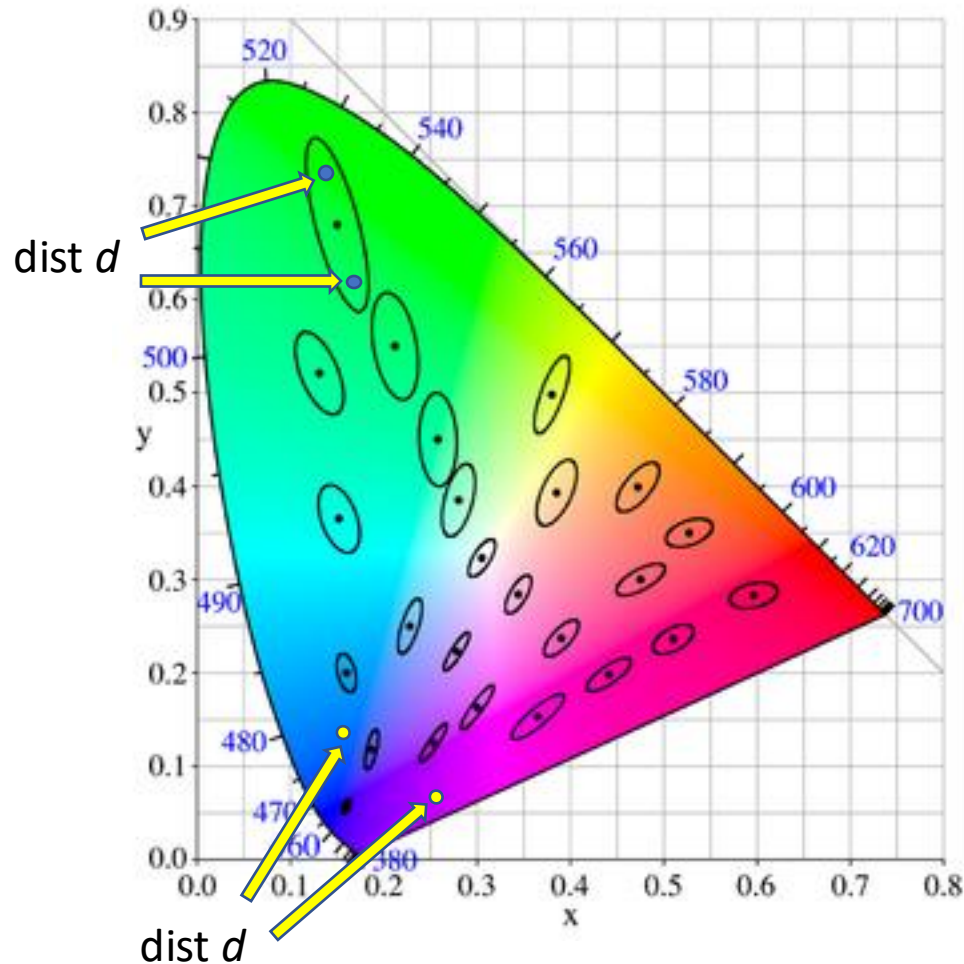
r-g coordinates

x-y coordinates

x-y Chromaticity Diagram



MacAdam Ellipses



- A MacAdam ellipsis is a region on a chromaticity diagram that contains all colors which are indistinguishable to the average human eye from the color at the center of the ellipse
- Trained observer does color matching experiment, repeatedly, for 25 different colors
- Matches are fit with an ellipse
- Shown here at 10X size
- We'd like to have some space where these are circles! And the *same size*.

https://en.wikipedia.org/wiki/MacAdam_ellipse

Questions: True or false?

- a) Because color matching operates as a linear system, if you know the trichromatic coefficients (r, g, b) of a color C_1 and of a color C_2 , then the trichromatic coefficients of the additive mixture of C_1 and C_2 will just be the sum of their separate trichromatic coefficients.
- b) Because color matching operates as a linear system, if you know the tristimulus values (R, G, B) of a color C_1 and of a color C_2 , then the tristimulus values of the additive mixture of C_1 and C_2 will just be the sum of their separate tristimulus values.
- c) MacAdam's ellipses are closer to being round and to being all the same size in the $u-v$ chromaticity diagram than in the $x-y$ chromaticity diagram.
- d) The color matching experiment was originally done using the XYZ primaries.
- e) If two colored lights C_1 and C_2 are metamers under scotopic (dim light) viewing conditions, then they will also be metamers under photopic (bright light) viewing conditions.
- f) If two colored lights C_1 and C_2 are metamers under photopic (bright light) viewing conditions, then they will also be metamers under scotopic (dim light) viewing conditions.
- g) If you know the trichromatic coefficients r, g, b of some color C_1 in the NTSC RGB color coordinate system, you can calculate the tristimulus values X, Y, Z , for that color in the CIE XYZ color coordinate system.

Last time

1. CIE RGB: (R_C, G_C, B_C) The monochromatic spectral primaries, used in the 1920's for the original color matching experiments
2. CIE XYZ: Used for color specification, all colors are in the positive quadrant
 - Reference system
3. For color display: RGB models
 - NTSC RGB
 - sRGB
4. Aiming at perceptual uniformity
 - UVW, $U^*V^*W^*$
 - Lab, $L^*a^*b^*$

Today

Systems for Color Image Processing

- Hue, Saturation, Intensity (HSI)
- Hue, Saturation, Value (HSV)
- Hue, Saturation, Lightness (HSL)

Gamma Correction

Systems for Color Compression & Transmission

- YIQ, YUV, YCrCb

Systems for Color Printing

- CMY, CMYK

Color Image Processing

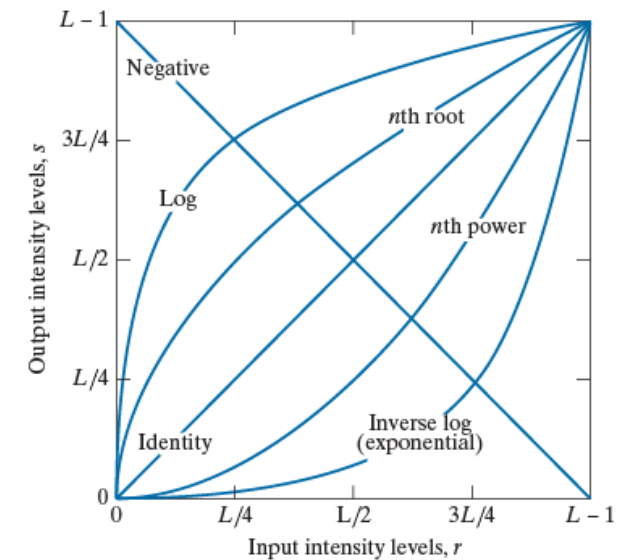
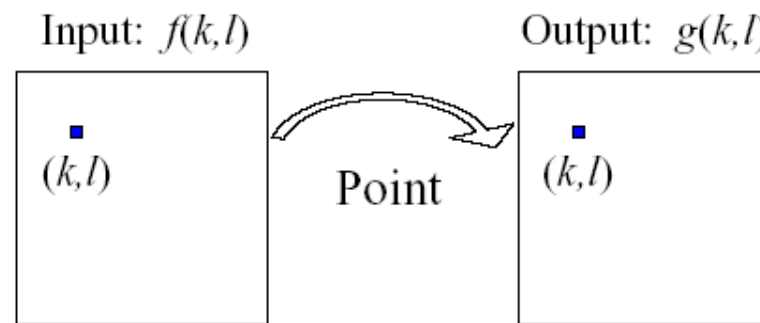
- Which color representation system to use for image processing?
- Do we have to modify a grayscale algorithm to be suitable for color images?

Let's consider an algorithm for grayscale image processing and see how it gets done for a color image

Histogram equalization = Popular method for contrast enhancement

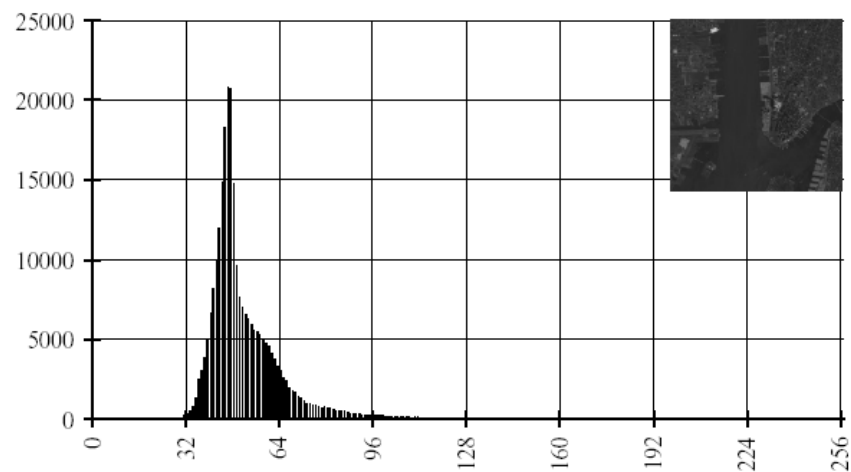
Let f = input gray level
Let g = output gray level

$$g = T(f)$$

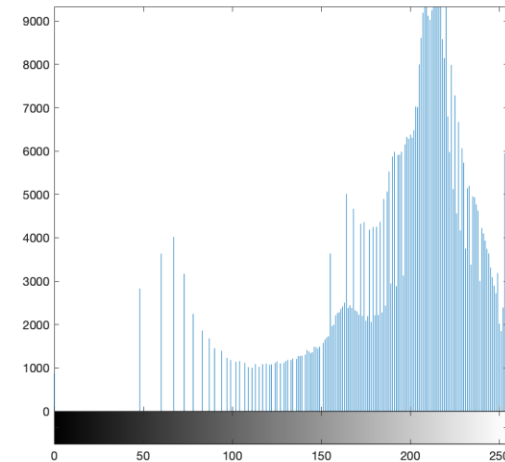




Too dark → make things brighter



Too bright → make things darker



Contrast enhancement (e.g., histogram equalization) in RGB??



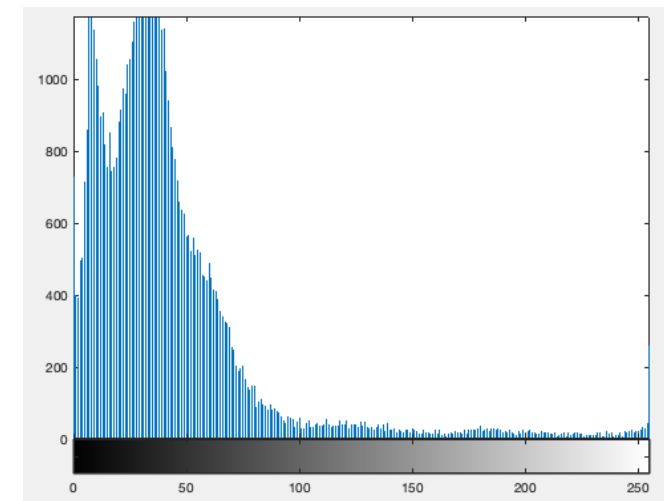
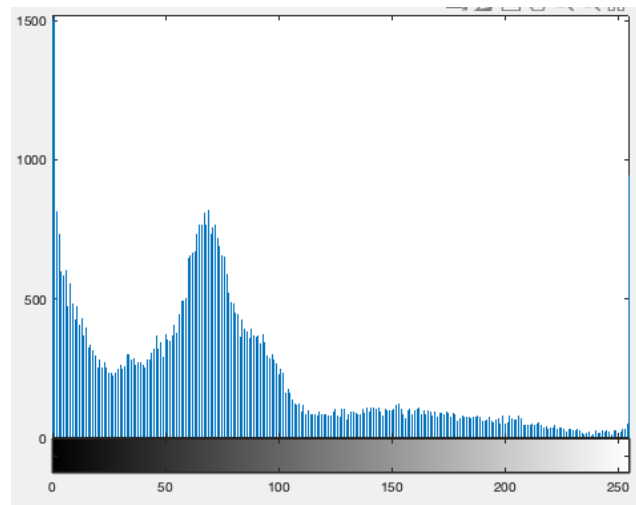
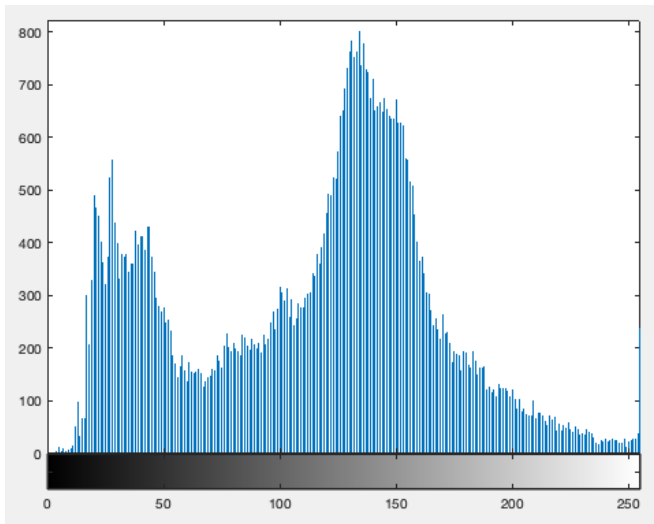
Red
mean = 114



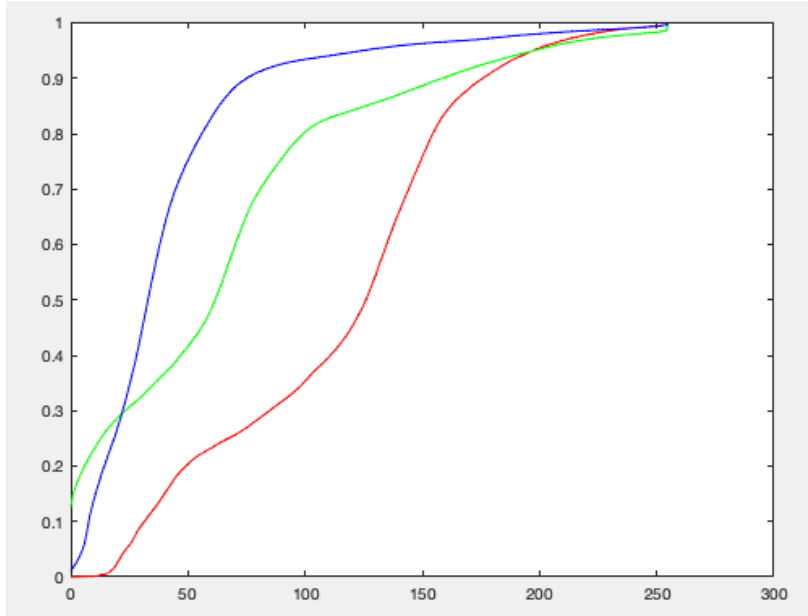
Green
mean = 67



Blue
mean = 43



Separate histogram equalization



Need to be able to manipulate the intensity component without (overly) disturbing the chromaticity (hue & saturation)

RGB $\rightarrow$ HSI

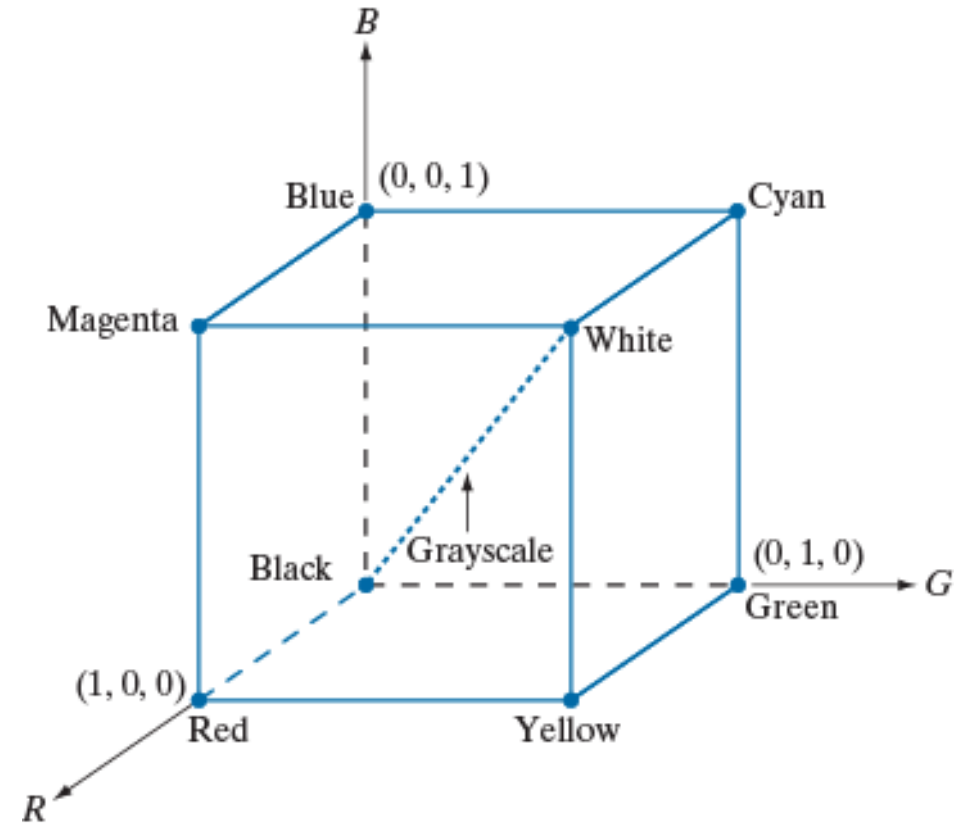
Color Cube

One way to think about displayable colors is with a color cube

Call the three primaries Red, Green, Blue

Black = (0,0,0), White=(1,1,1)

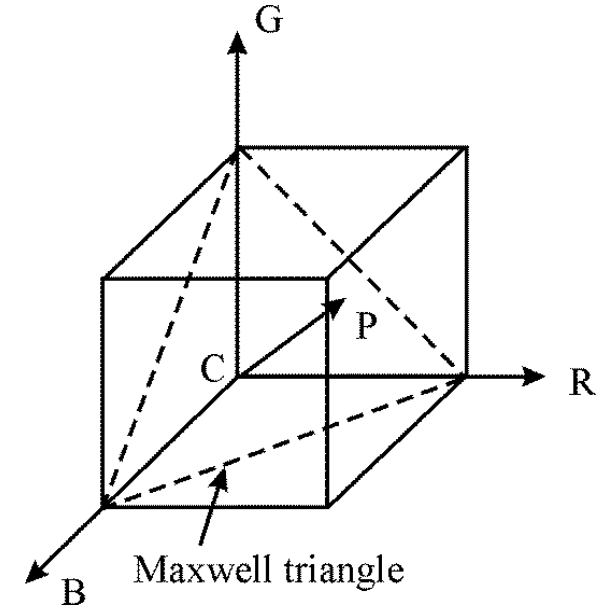
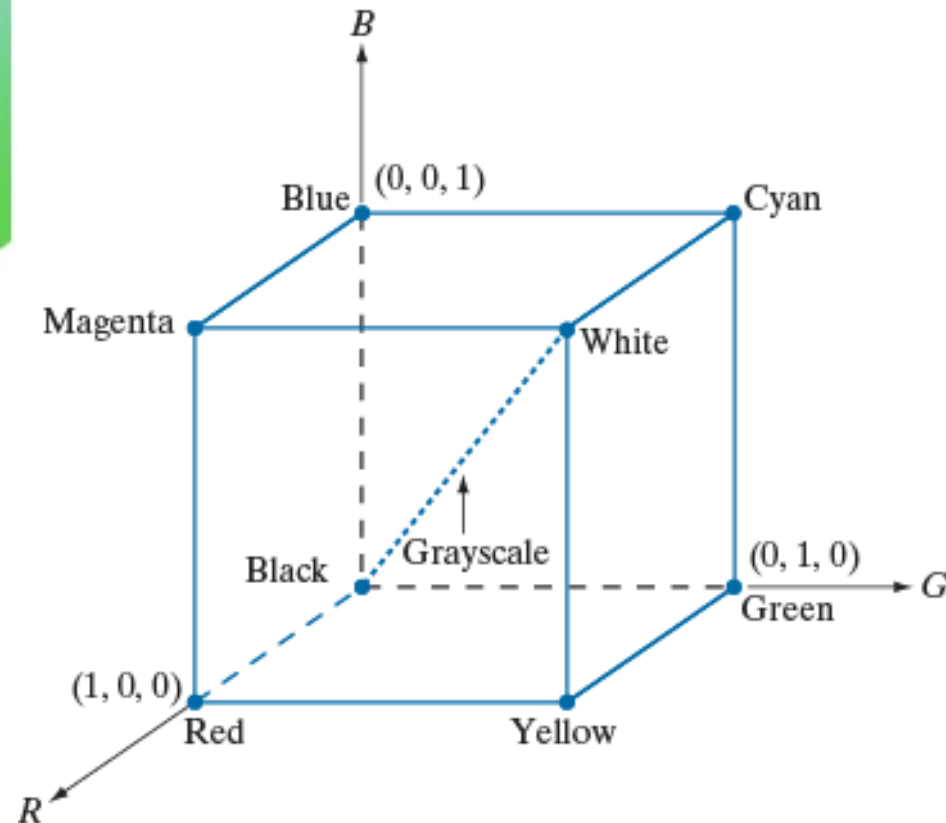
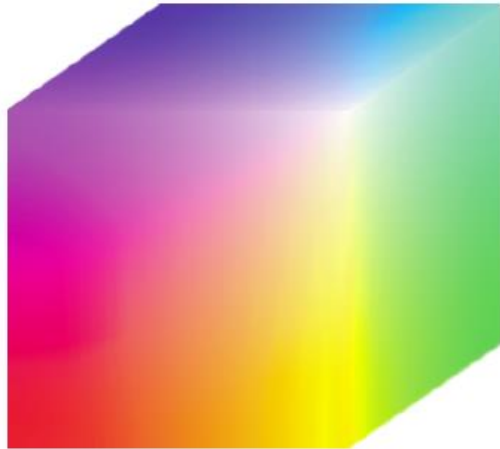
No negative tristimulus values



Maxwell triangle inside the Color Cube

Black = (0,0,0),
White=(1,1,1)

For purposes of this discussion,
 $0 \leq R, G, B \leq 1$
 $0 \leq H, S, I \leq 1$



The Maxwell Triangle is inside
the color cube

Lies in plane: $R + B + G = 1$

Has vertices at (1,0,0), (0,1,0) &
(0,0,1)

Maxwell Triangle

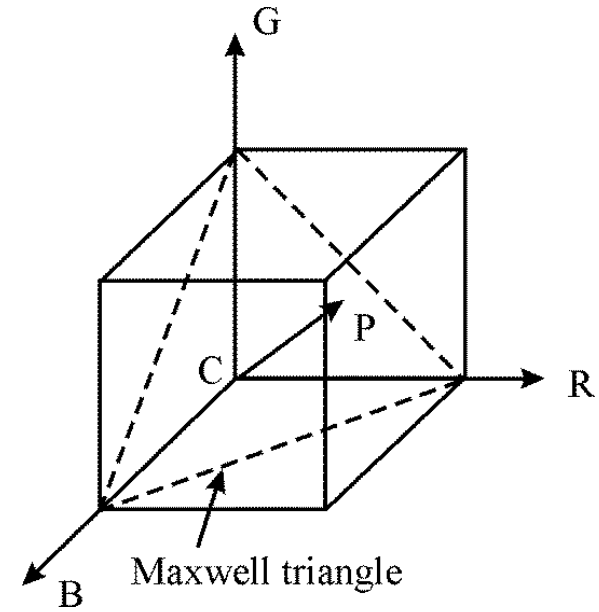
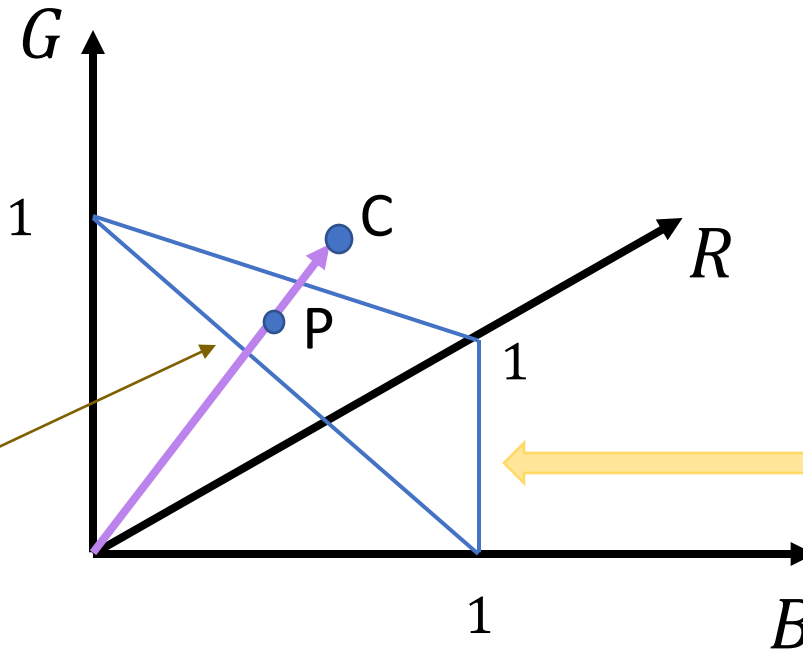
Lies in plane: $R + B + G = 1$

Has vertices at $(1,0,0)$, $(0,1,0)$ & $(0,0,1)$

Let C be some color with:

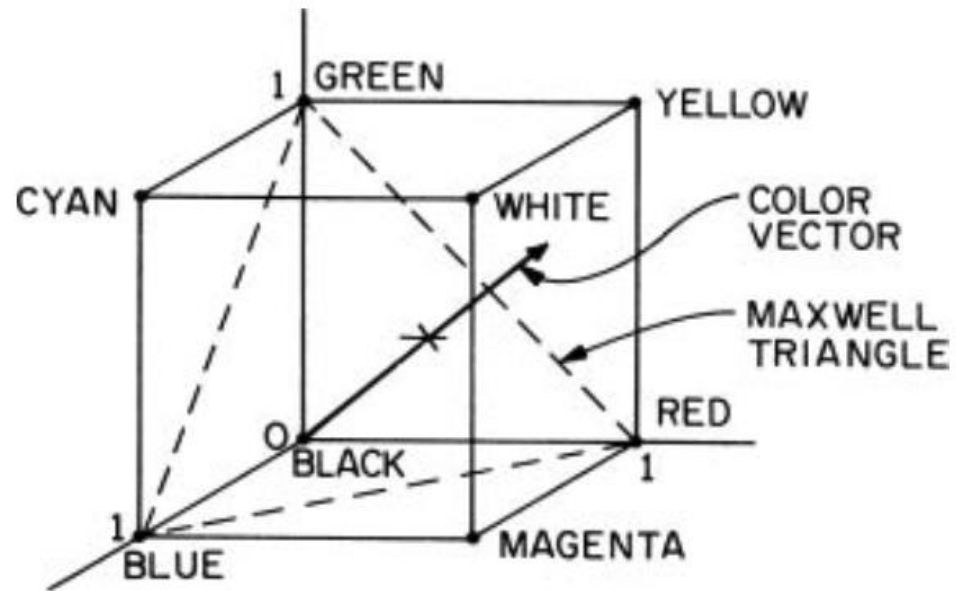
- tristimulus values R', G', B'
- trichromatic coefficients r', g', b'

This point P where color vector C intersects the Maxwell triangle has coordinates r', g', b'



Remember colors C , $2C$, $0.5C$ have different tristimulus values but the same trichromatic coefficients

HSI Color Model: Intensity



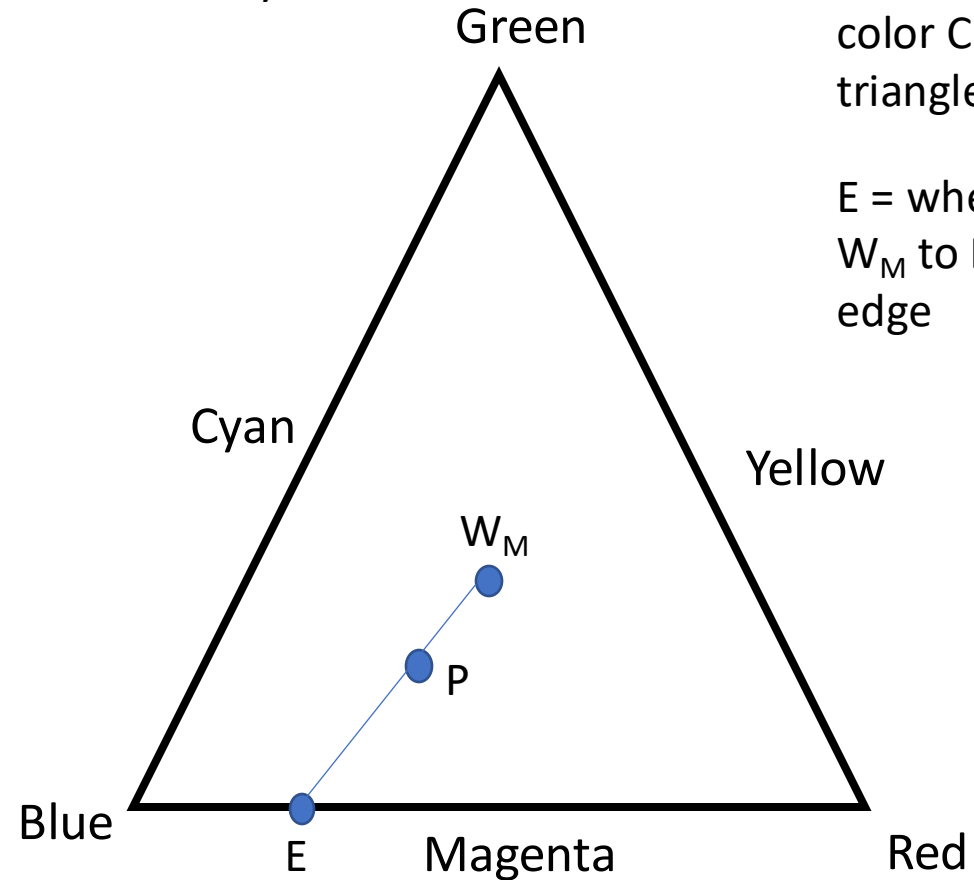
$$I = \frac{1}{3}(R + G + B)$$

H = Hue
S = Saturation
I = Intensity

W_M = center of triangle

P = point where our color C intersects the triangle

E = where the line from W_M to P hits the triangle edge



Maxwell Triangle

Color Planes



R(j,k)



G(j,k)



B(j,k)

$$I = \frac{1}{3}(R + G + B)$$

$$I(j, k) = \frac{1}{3}(R(j, k) + G(j, k) + B(j, k))$$

HSI Color Model - Saturation

$$\text{Saturation} = S = \frac{|W_M P|}{|W_M E|} = 1 - \frac{3\min(R, G, B)}{R + G + B}$$

$|W_M E|$ means length of line segment from W_M to E

$$S(j, k) = 1 - \frac{3\min[R(j, k), G(j, k), B(j, k)]}{R(j, k) + G(j, k) + B(j, k)}$$

If $I = 0$, then S is undefined

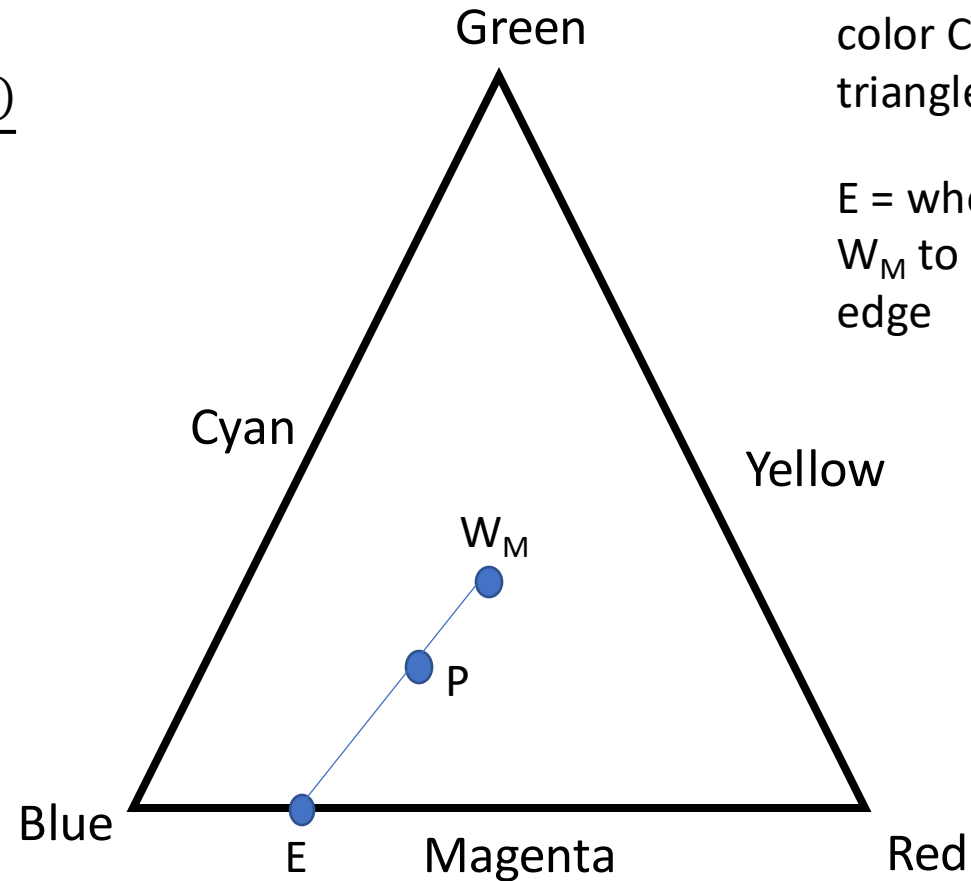
As $P \rightarrow W_M$, $S \rightarrow 0$

As $P \rightarrow$ triangle edge E , $S \rightarrow 1$

W_M = center of triangle

P = point where our color C intersects the triangle

E = where the line from W_M to P hits the triangle edge



Maxwell Triangle

HSI Saturation

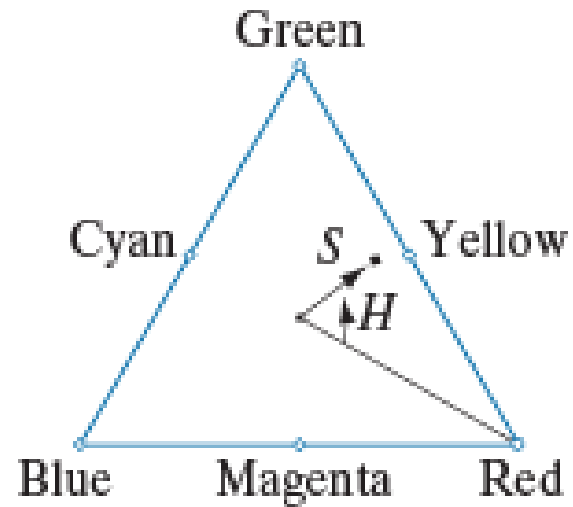
$$S = 1 - \frac{3\min(R, G, B)}{R + G + B}$$

Red has $(R, G, B) = (1, 0, 0)$

What is the saturation of Red?

Yellow has $(R, G, B) = (1, 1, 0)$

What is the saturation of Yellow?

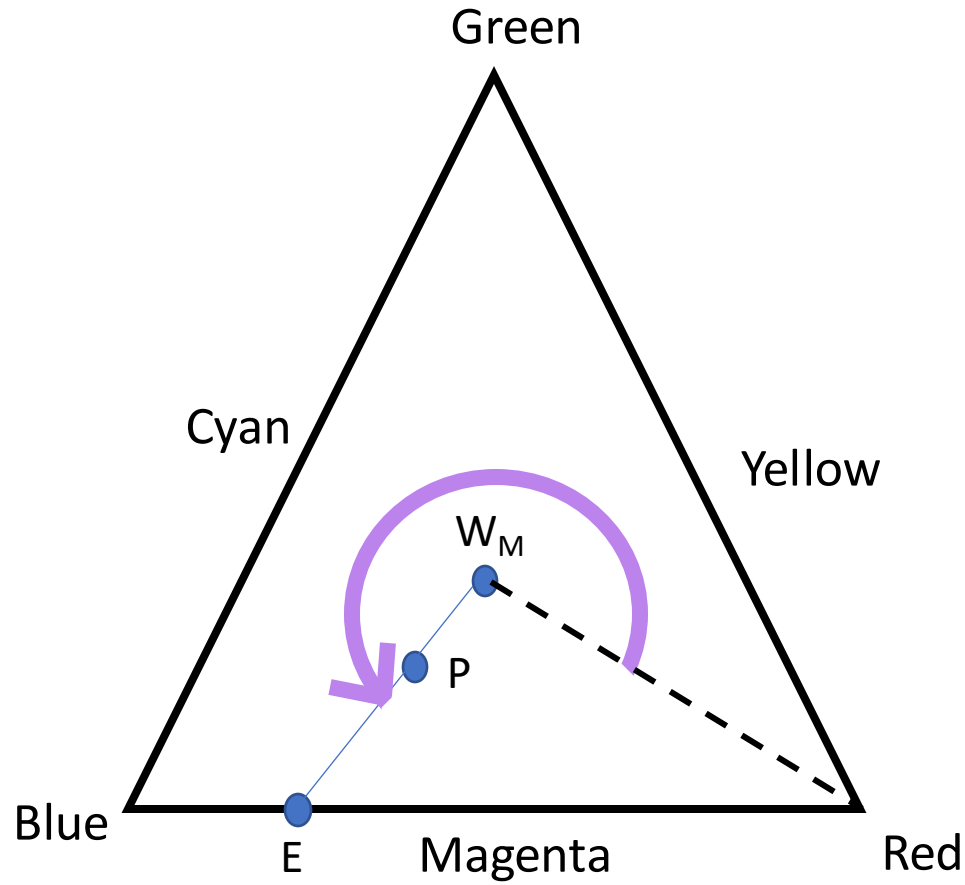


HSI Color Model - Hue

W_M = center of triangle

P = point where our color C intersects the triangle

E = where the line from W_M to P hits the triangle edge



$$\text{Hue} = H = \frac{\text{angle from Red}}{2\pi}$$

$$0 \leq H \leq 1$$

If $S = 0$, P & W_M are the same point, and H is undefined

If $I = 0$, then H is undefined

HSI Question 1

Which plane is H, S, I ??



A



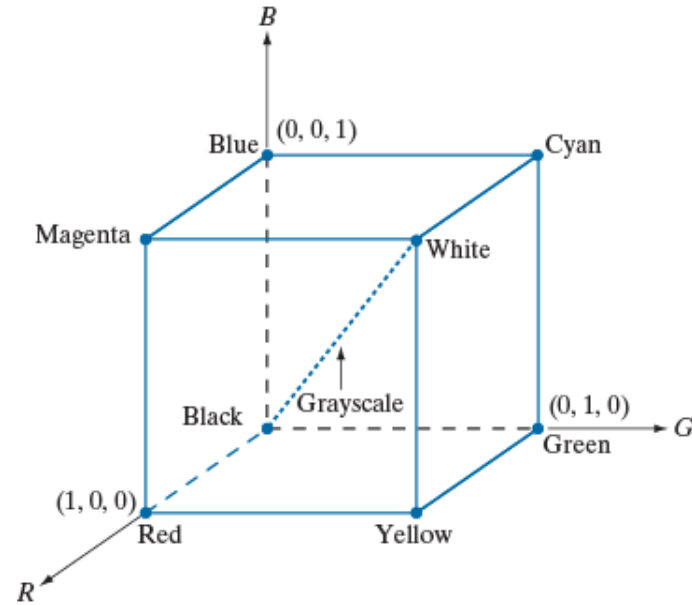
B



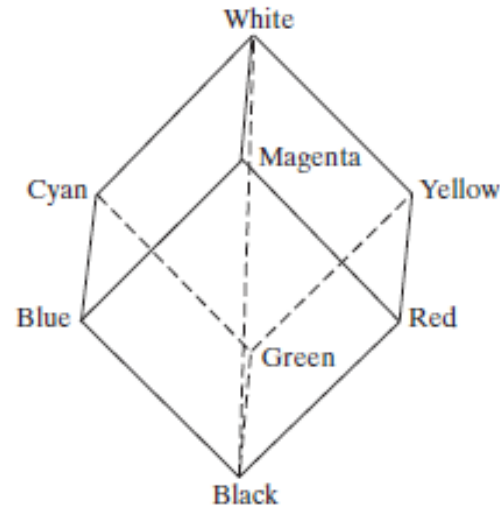
C



Another way of visualizing the HSI model

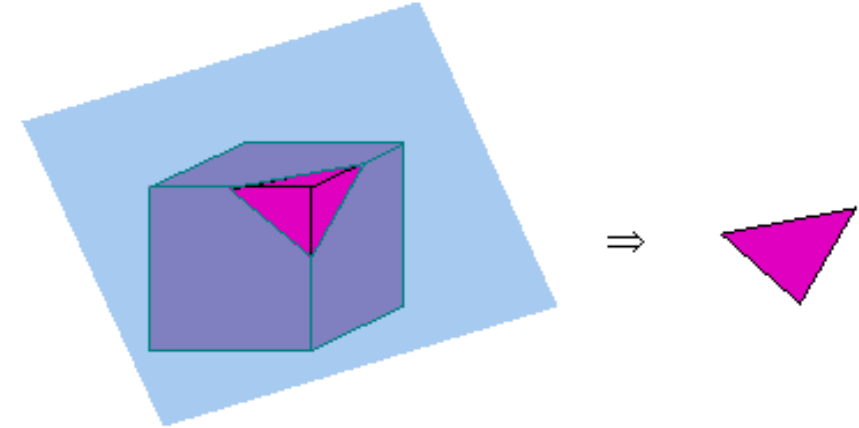
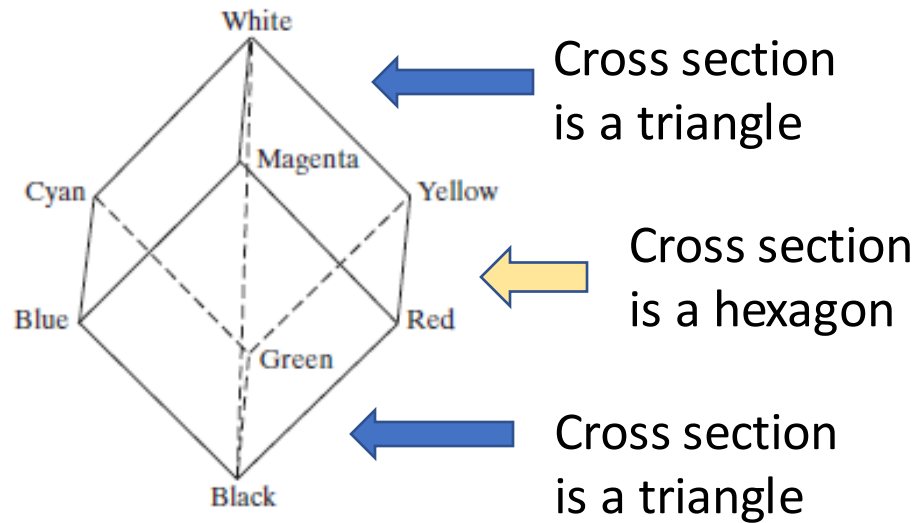


RGB cube



- Stand the cube on its black vertex
- Intensity (grayscale) axis runs from Black to White
- For a color C, take plane through C perpendicular to Black-White axis, see where it intersects axis
- That gives you the intensity

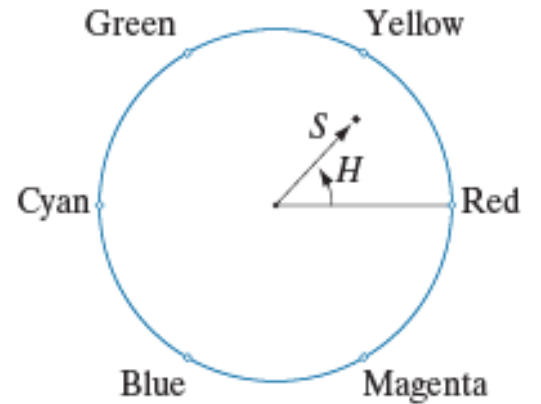
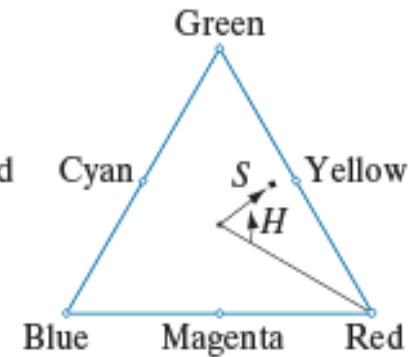
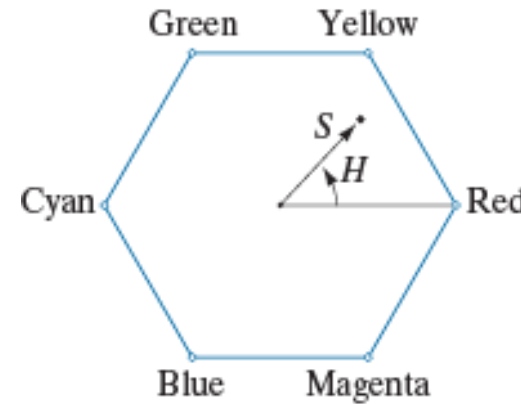
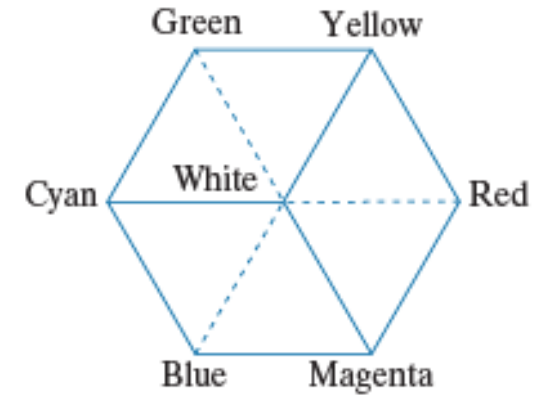
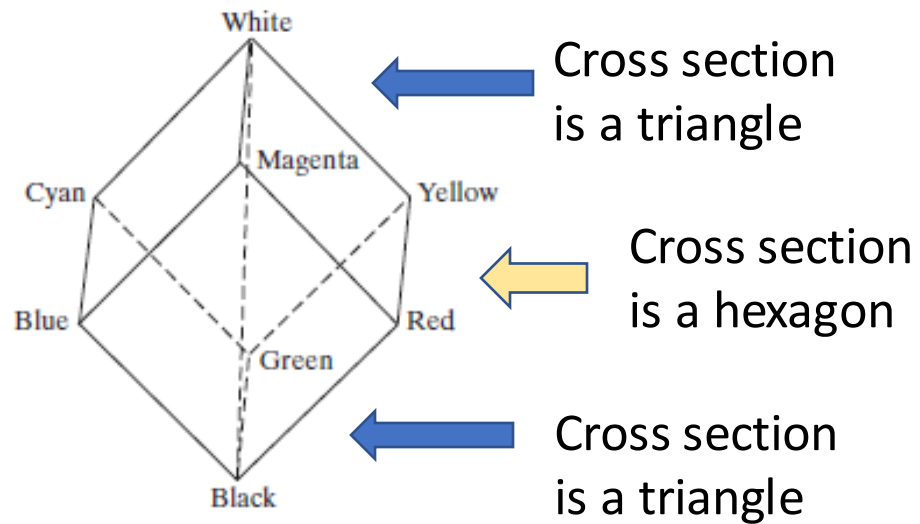
Another way of visualizing the HSI model



From: <https://old.mathleague.com/help/geometry/3space.htm>



Various ways of visualizing the HSI model



Gonzalez & Woods,
Digital Image
Processing, Fig. 7.11

HSI Question 2

$$I = \frac{1}{3}(R + G + B)$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

If $S = 0$ or $I = 0$, then H is undefined

If $I = 0$ then S is undefined

$$S = 1 - \frac{3\min(R, G, B)}{R + G + B} \quad \text{+}\epsilon$$

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{\frac{1}{2}}} \right\} \quad \text{+}\epsilon$$

Question: Suppose we start with a color $C = (R, G, B)$ where $0 < R, G, B \leq 0.5$

We multiply the values by 2 (staying within the RGB cube) to get $C' = (R', G', B') = (2R, 2G, 2B)$.

How are the (H, S, I) values changed?

HSI Answer (Equation Crunching)

$$I' = \frac{1}{3}(R' + G' + B')' = \frac{1}{3}(2R + 2G + 2B) = 2I$$

$$S' = 1 - \frac{3 \min(R', G', B')}{R' + G' + B'} = 1 - \frac{3 \min(2R, 2G, 2B)}{2R + 2G + 2B} = S$$

$$H' = \begin{cases} \theta' & \text{if } 2B \leq 2G \\ 360 - \theta' & \text{if } 2B > 2G \end{cases} \quad \theta' = \cos^{-1} \left\{ \frac{\frac{1}{2}[(2R - 2G) + (2R - 2B)]}{[(2R - 2G)^2 + (2R - 2B)(2G - 2B)]^{\frac{1}{2}}} \right\} = \theta$$

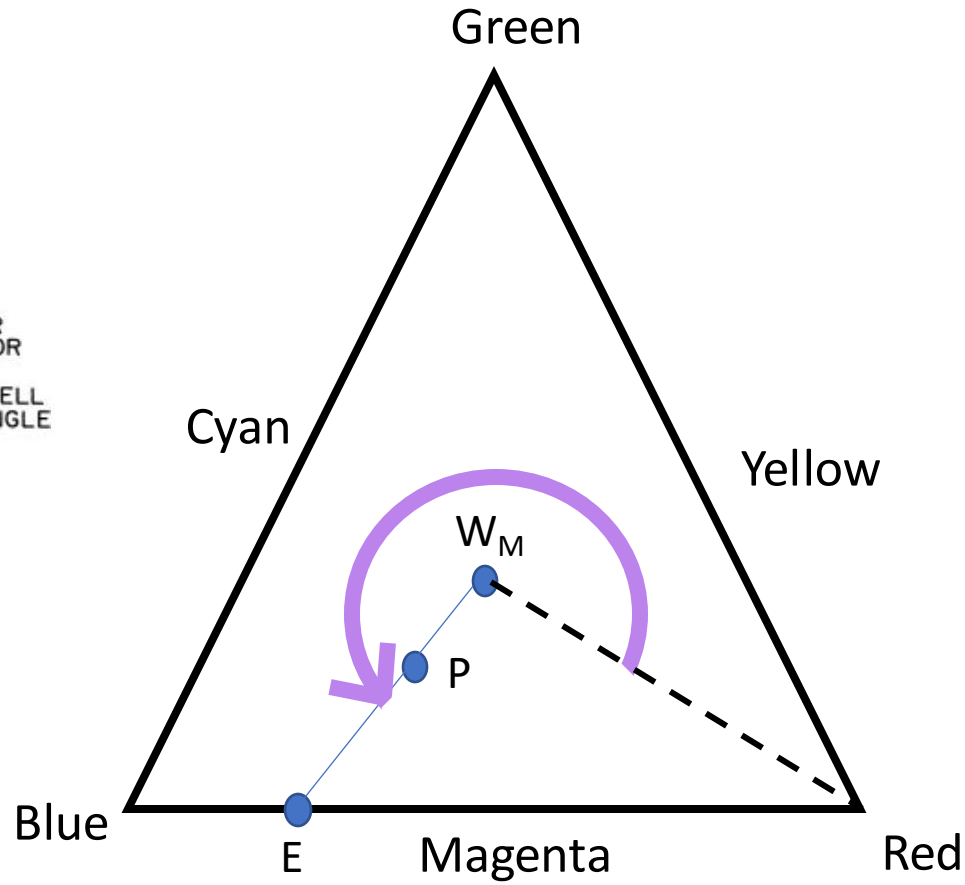
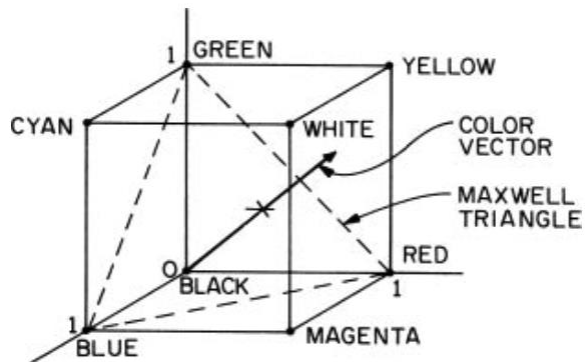
$C = (R, G, B)$ has coordinates (H, S, I)

$C' = (R', G', B') = (2R, 2G, 2B)$

C' has coordinates $(H, S, 2I)$

HSI Answer (Maxwell Triangle Visualization)

$$I = \frac{1}{3}(R + G + B)$$



$C = (R, G, B)$ has coordinates (H, S, I)

$$C' = (R', G', B') = (2R, 2G, 2B)$$

$$\text{Saturation} = S = \frac{|W_M P|}{|W_M E|}$$

$$\text{Hue} = H = \frac{\text{angle from Red}}{2\pi}$$

Color $C' = 2C$ intersects the triangle at the same point P

Conclusion: there's no change to hue or saturation

C' has coordinates $(H, S, 2I)$

Coordinate for Intensity or Brightness or Lightness

$$I = \frac{1}{3}(R + G + B) \quad \text{Intensity}$$

$$Y = 0.299R + 0.587G + 0.114B \quad \text{Luma, considered to be a better perceptual weighting than } 1/3, 1/3, 1/3$$

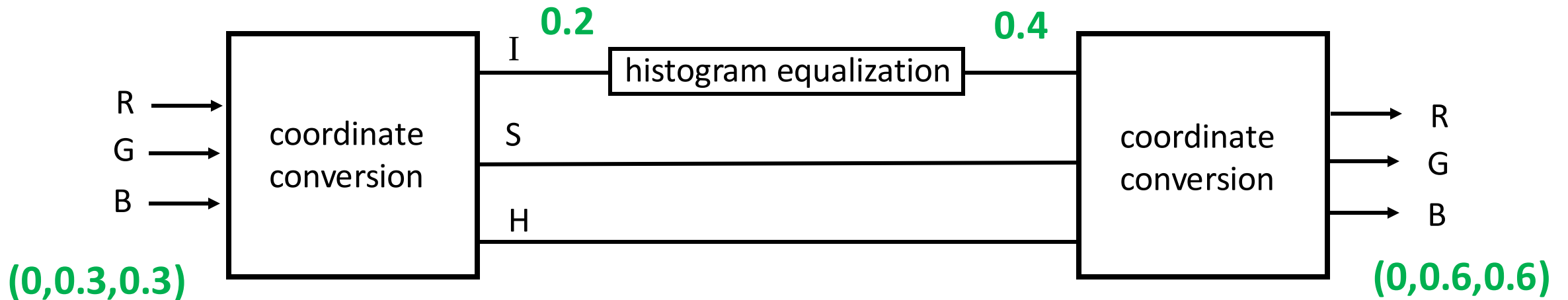
$$V = \max(R, G, B) \quad \text{Value}$$

$$L = \frac{\max(R, G, B) + \min(R, G, B)}{2} \quad \text{Lightness}$$

Many systems: HSI, HSV, HSL, etc.

Intensity Processing

In principle, can do histogram equalization, contrast stretching, anything you would do on a grayscale image



Color C: $(R,G,B) = (0, 0.3, 0.3)$

Intensity = 0.2

Make CDF, suppose equalized intensity = 0.4

Hue and Saturation are unchanged

Easy case: New RGB values are: $(0, 0.6, 0.6)$

Histogram Equalization on Intensity (Problem case)

Color C: (R,G,B) = (0.8, 0.2, 0.2)

Intensity = 0.4

Suppose equalized intensity = 0.8

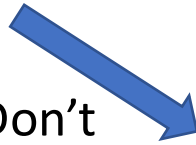
New RGB values would be (1.6, 0.4, 0.4)

1. Truncate



(1.0, 0.4, 0.4)

2. Don't
equalize
fully



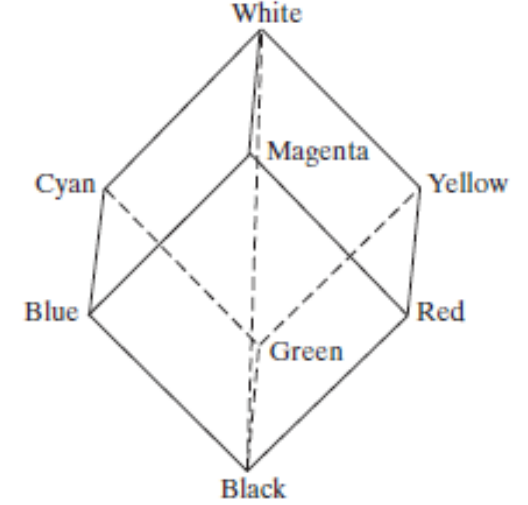
(1.0, 0.25, 0.25)



3. Something else

(1.0, 0.7, 0.7) ?

(1.0, 0.3, 0.3) ?

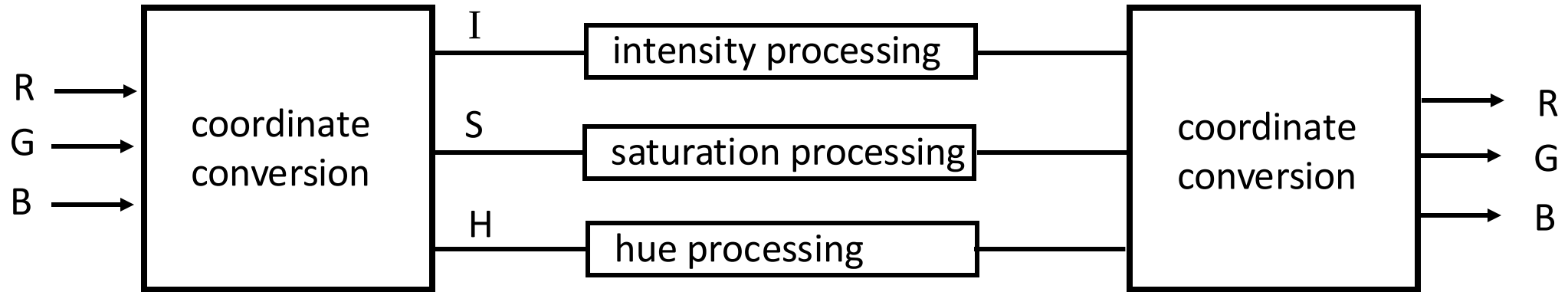


What has changed?

- Intensity?
- Saturation?
- Hue?

Color Image Processing

A system like HSI decouples the Intensity, Hue, and Saturation information



Saturation Adjustment

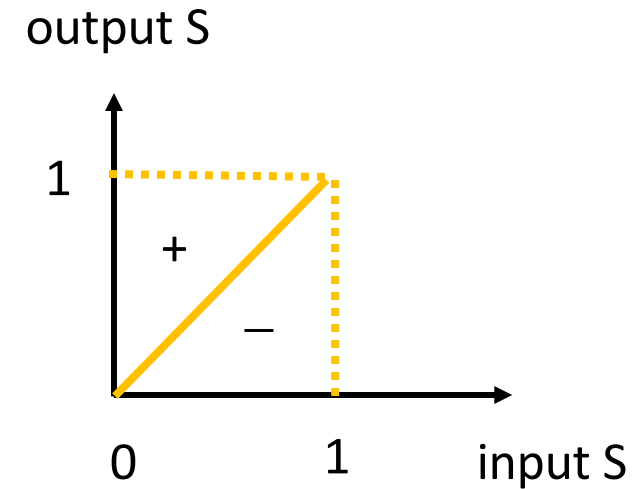


Adjusting saturation for artistic effect

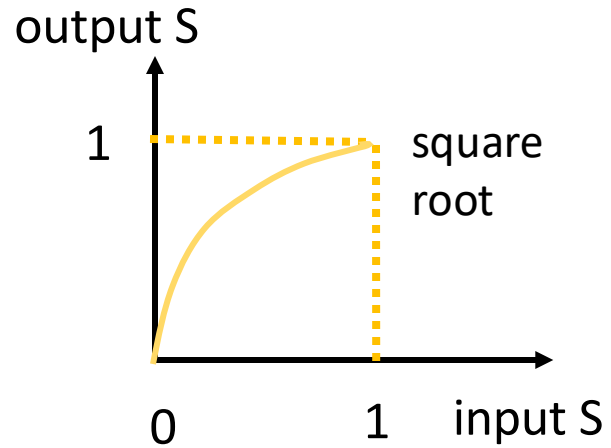


Saturation Processing

1. Often want to boost saturation (make things more colorful, richer)
2. Sometimes want to decrease saturation (e.g., make things look sad or mysterious)
3. Can do this in a selective way:
 - Boost or decrease saturation for particular regions of the image
 - Boost or decrease saturation for particular hues
 - Boost or decrease saturation for particular saturation levels



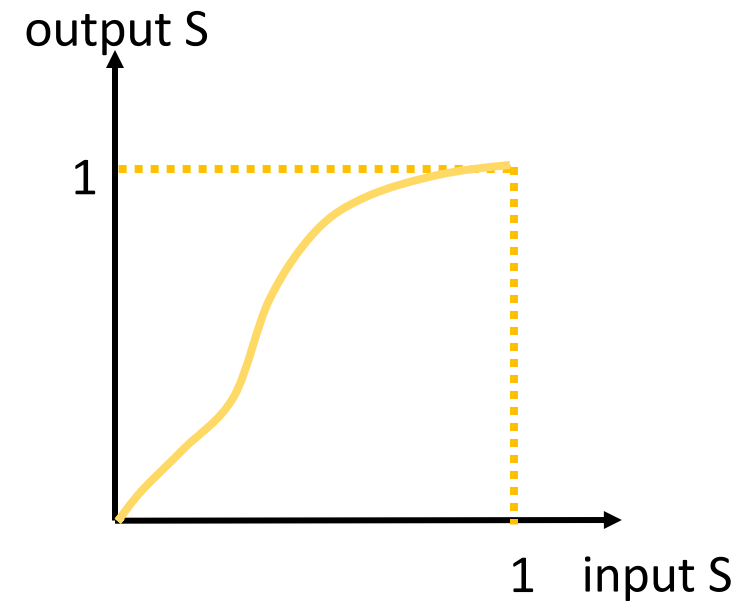
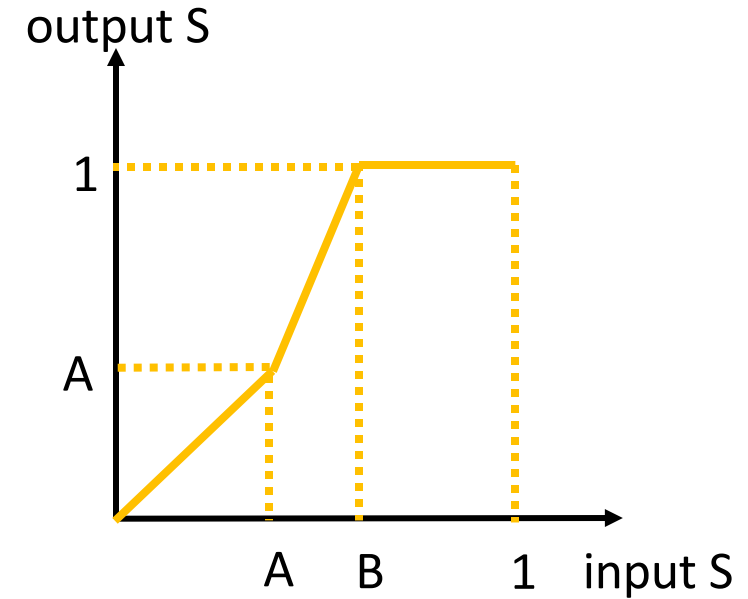
Saturation Processing



Boost saturation with
 $\text{newS} = \text{sqrt}(S) ??$

Beware of large
boosts to colors with
near-zero saturation!

Achromatic areas may
suddenly take on an
“arbitrary” hue





Original



Saturation enhancement with square root



Histogram equalization

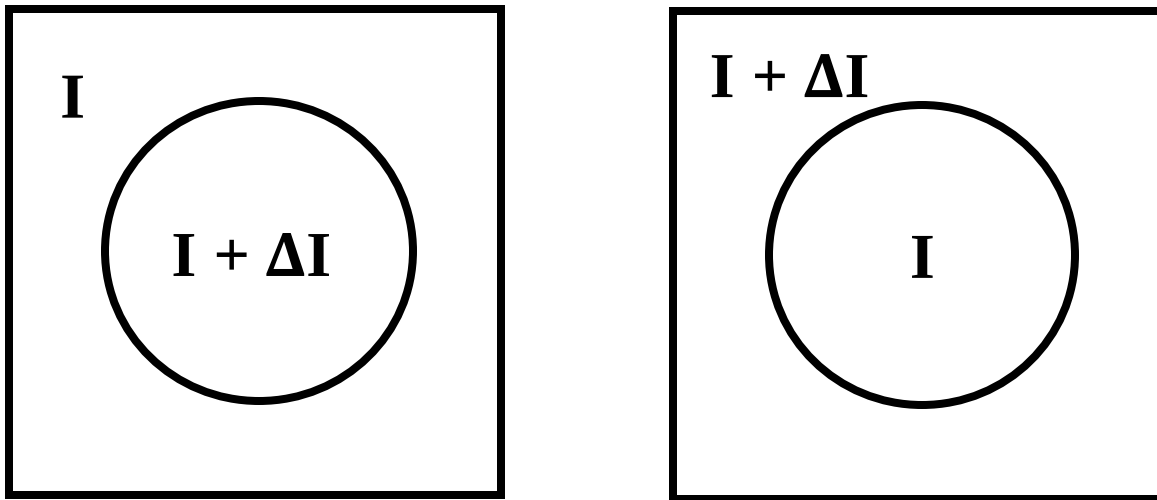


Less saturation enhancement for near whites

Weber's Law - Increment Sensitivity - Gamma Correction

Experimenter chooses an intensity I

Observer is presented with one of two patterns



Observer is asked “Is background or circle brighter?”

Keep I fixed, try different values of ΔI

Two alternative forced choice experiment

$\Delta I \rightarrow 0$, subject guesses randomly, 50% right

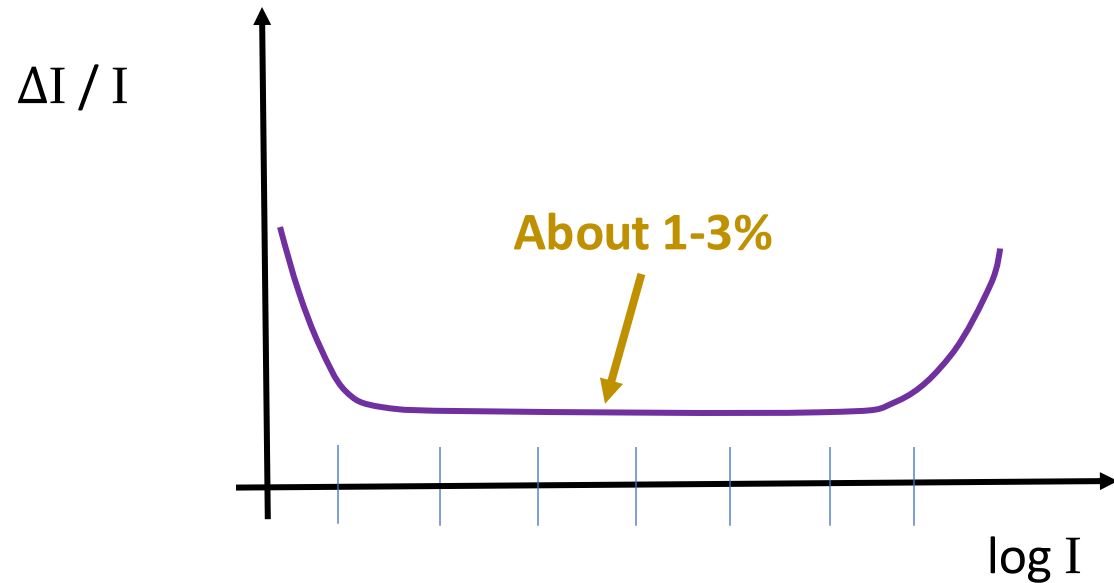
$\Delta I \rightarrow \text{Large}$, subject gets 100% right

Find ΔI so you get target value, say 75% correct

For the intensity I , this value of ΔI is called the just-noticeable difference (jnd)

Increment sensitivity

Weber's Law



$$\frac{\Delta I}{I} \approx \text{constant} \approx 0.02$$

Assumes you are properly light adapted

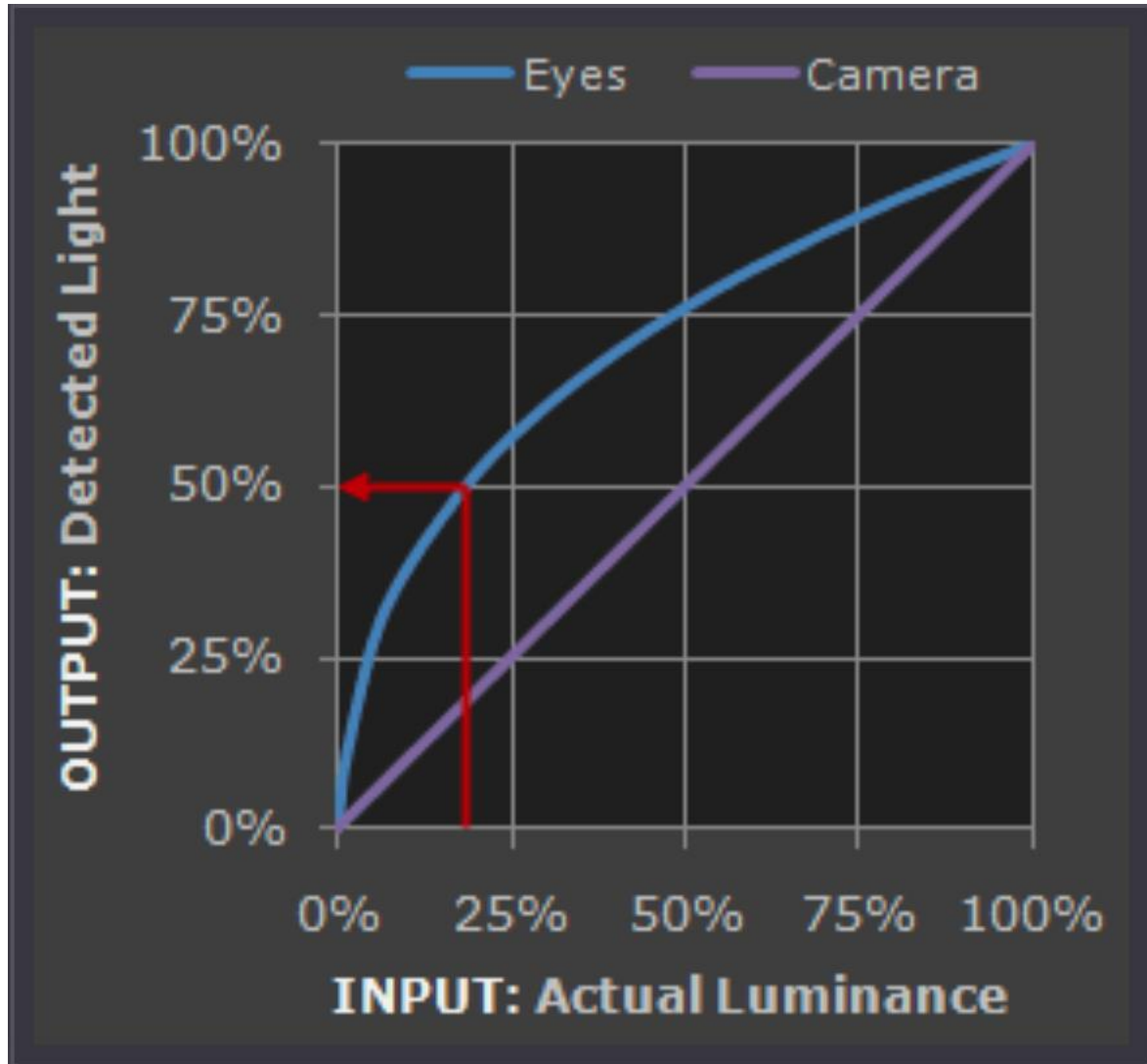
Similar effects elsewhere:

- Weight
- Price (behavioral economics)

Implications for image processing?

- Small differences in intensity are more noticeable in a dark region than in a bright region
- Image compression: Can allow more error in bright areas
- But this effect is often already encoded in the pixel values.... called gamma encoding

Gamma Encoding - Perceptual capture of images



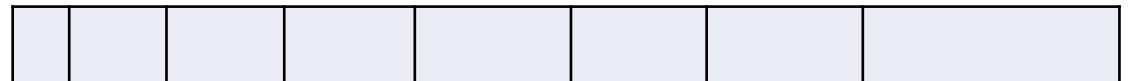
Range of gray levels....



Suppose we quantize it to 3 bits (8 possible gray levels)

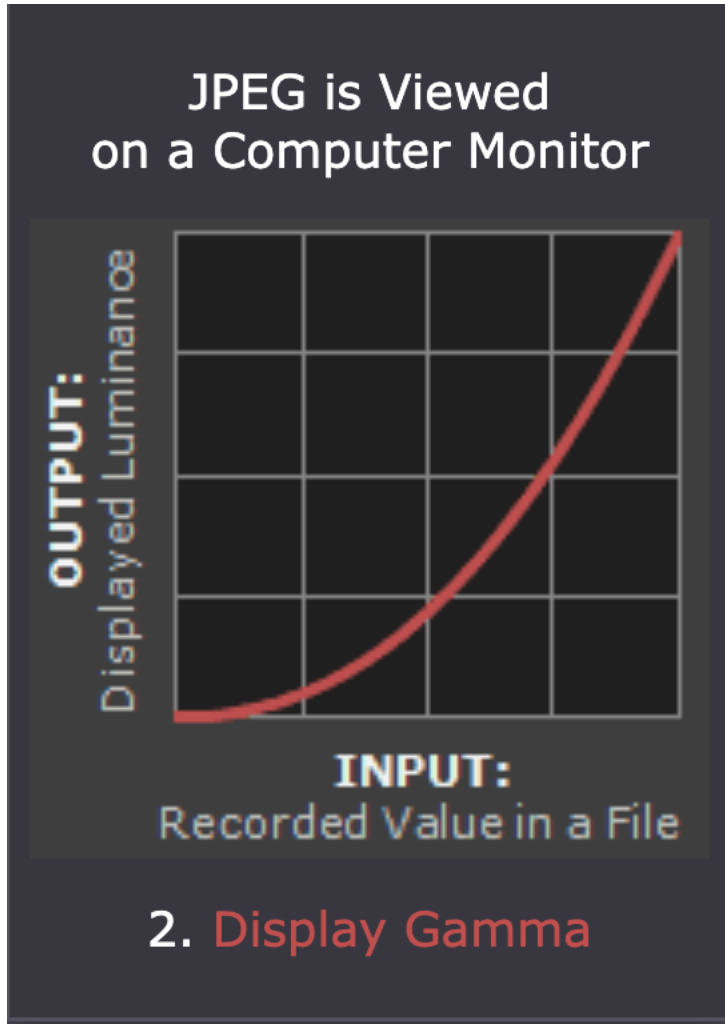
1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

- Quantization buckets are uniform (in photons)
- Quantization buckets are non-uniform (perceptually)



- Buckets are uniform (by human perception)
- Non-uniform (in terms of camera perception)

Gamma Encoding - Gamma Correction



Monitors typically have a nonlinear relationship between the applied voltage and the displayed luminance:

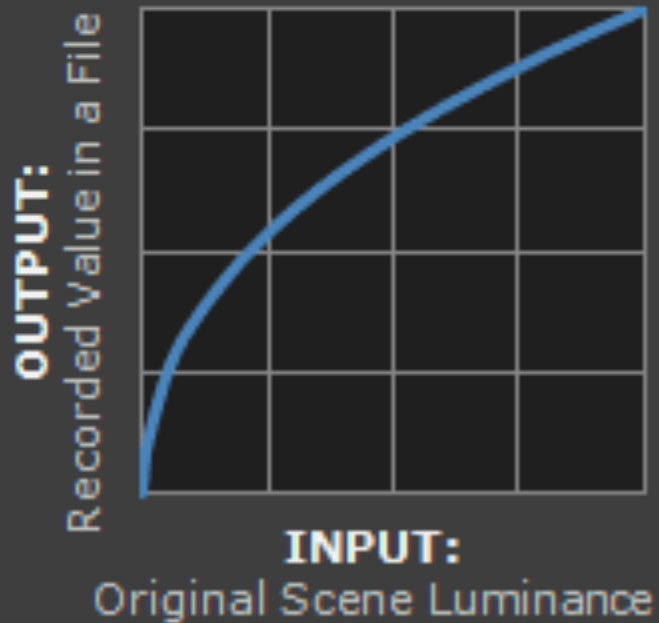
$$L \propto c v^\gamma$$

Typical values are $1.8 < \gamma < 2.5$

The overall processing chain might include several different gamma corrections.

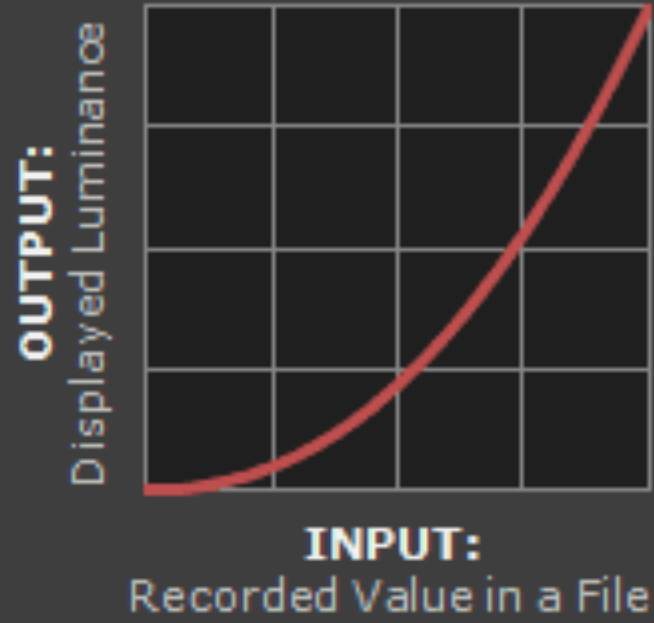
Gamma Encoding - Gamma Correction (Ideal)

RAW Camera Image
is Saved as a JPEG File



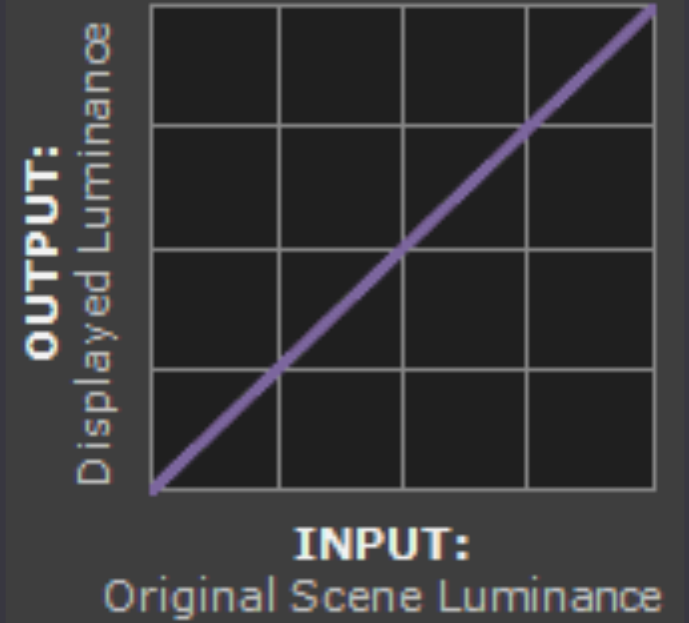
1. Image File Gamma

JPEG is Viewed
on a Computer Monitor



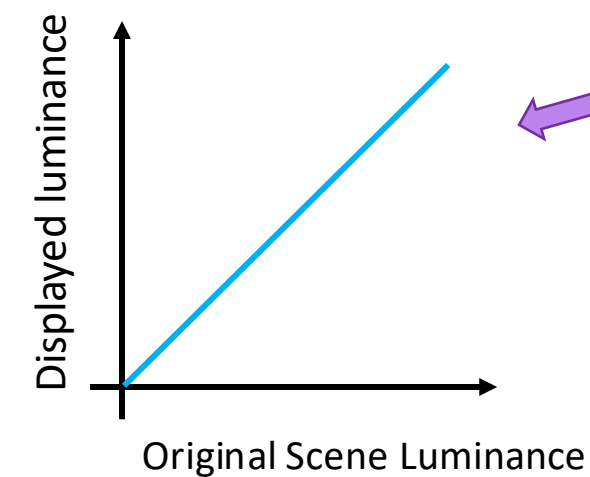
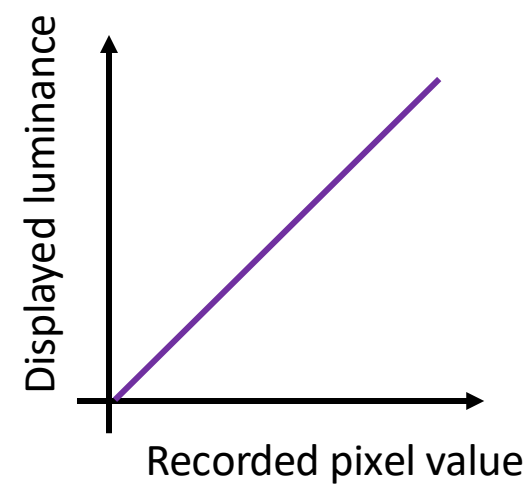
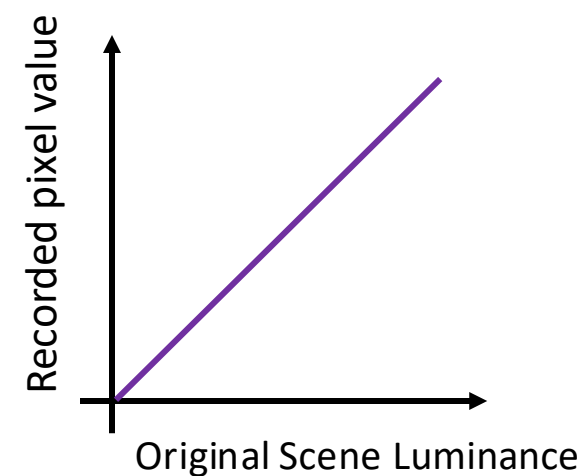
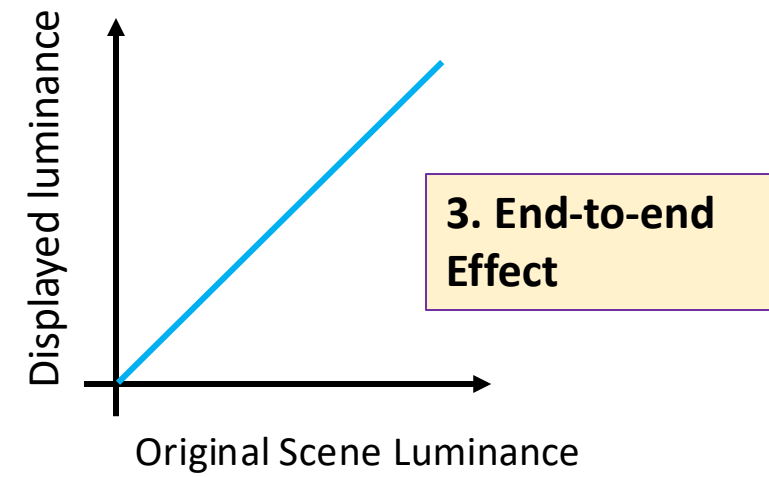
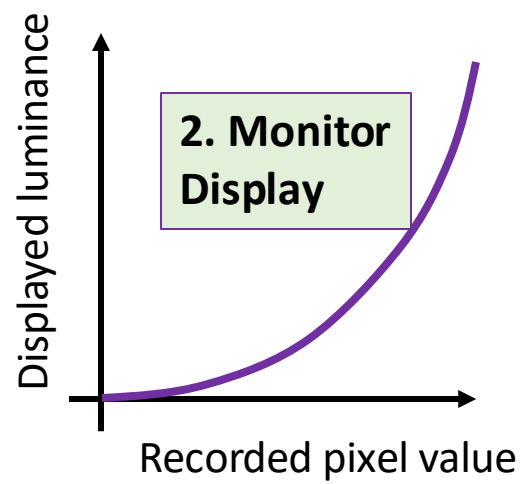
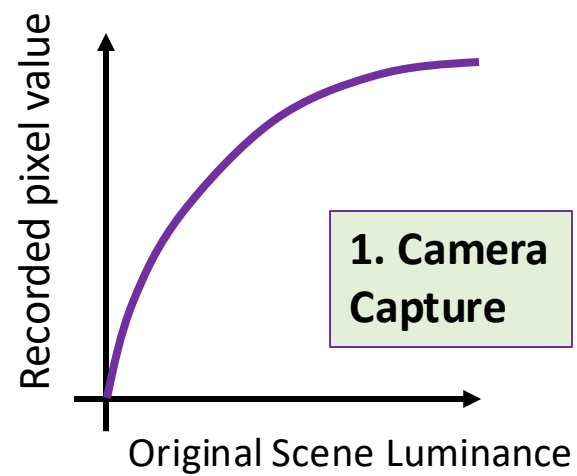
2. Display Gamma

Net Effect



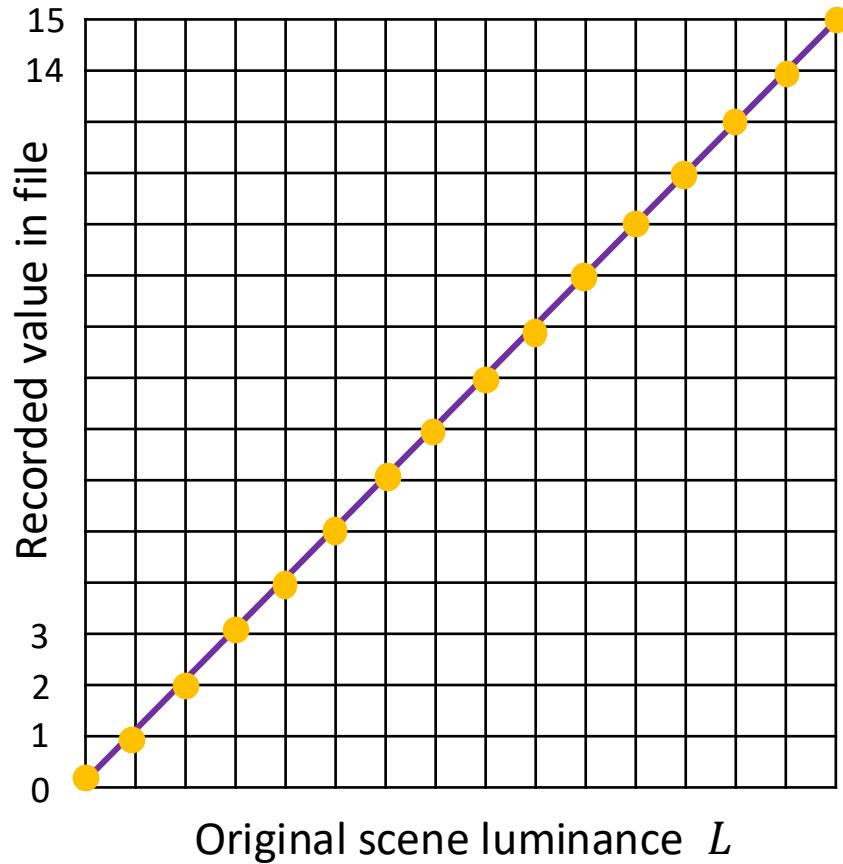
3. System Gamma

Gamma Correction

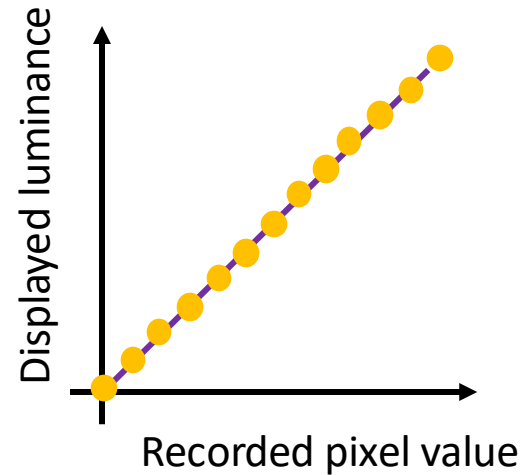


It's a straight line either way, so why bother with the nonlinearity?

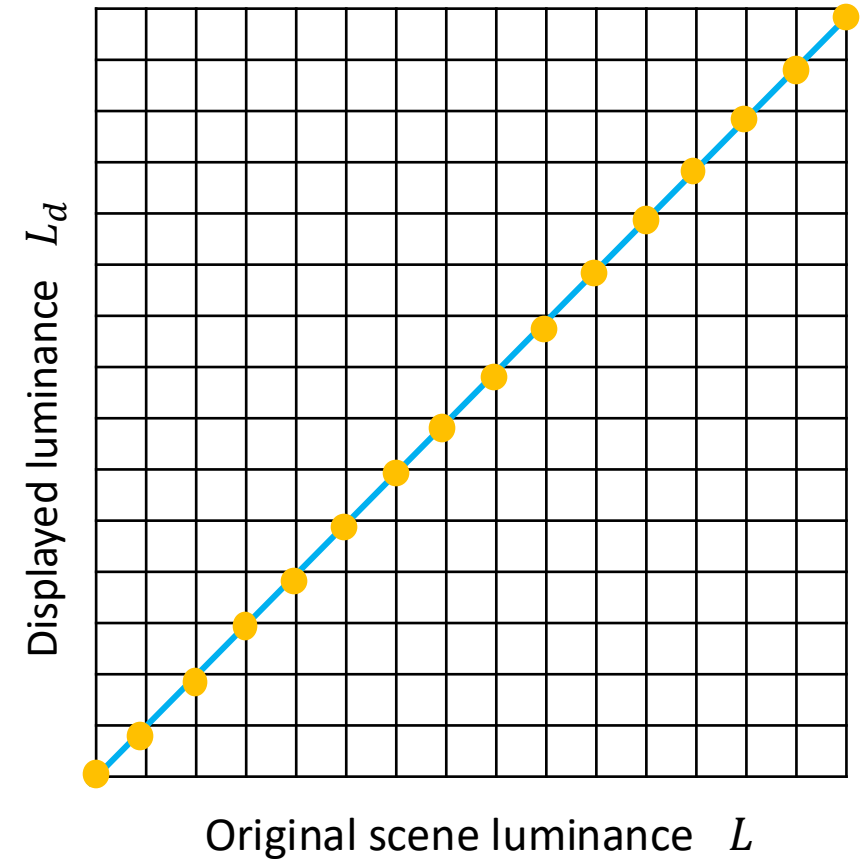
Luminance Encoding - Linear (example with 16 levels)



1. Camera Capture



2. Monitor Display



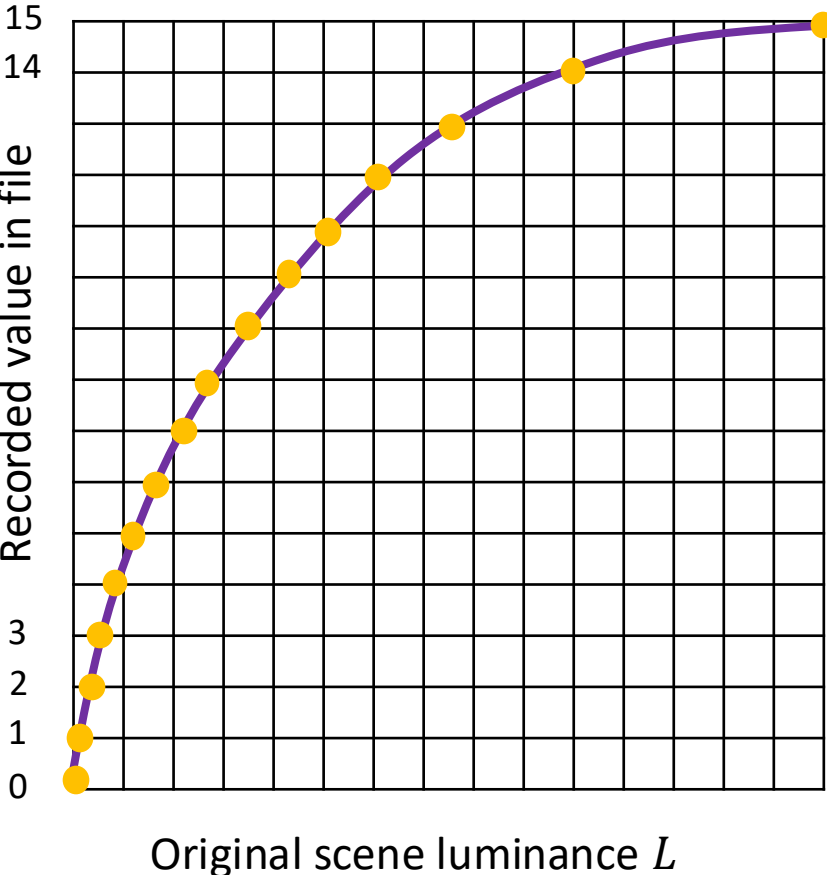
$$L_d = L$$

3. End-to-end Effect

Gamma Encoding - Nonlinear

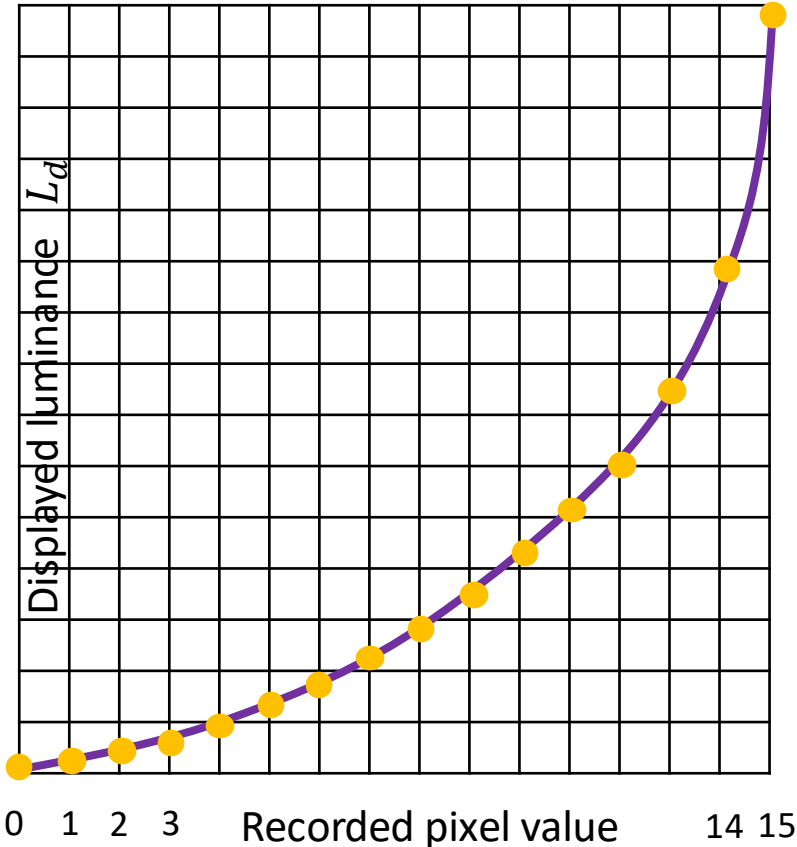
1. Camera Capture

$$v \propto L^{1/\gamma}$$

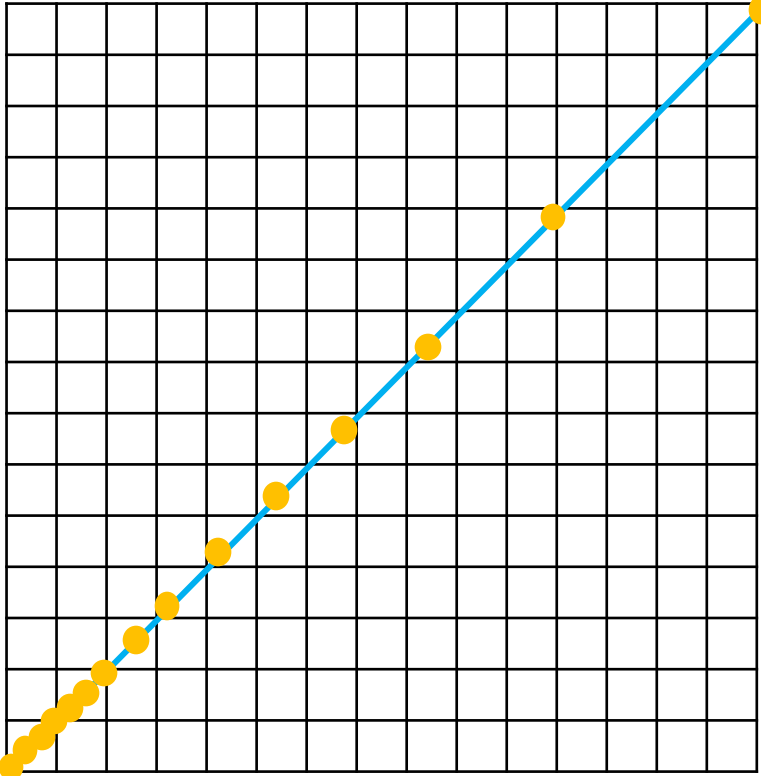


2. Monitor Display

$$L_d \propto v^\gamma$$



Displayed luminance L_d

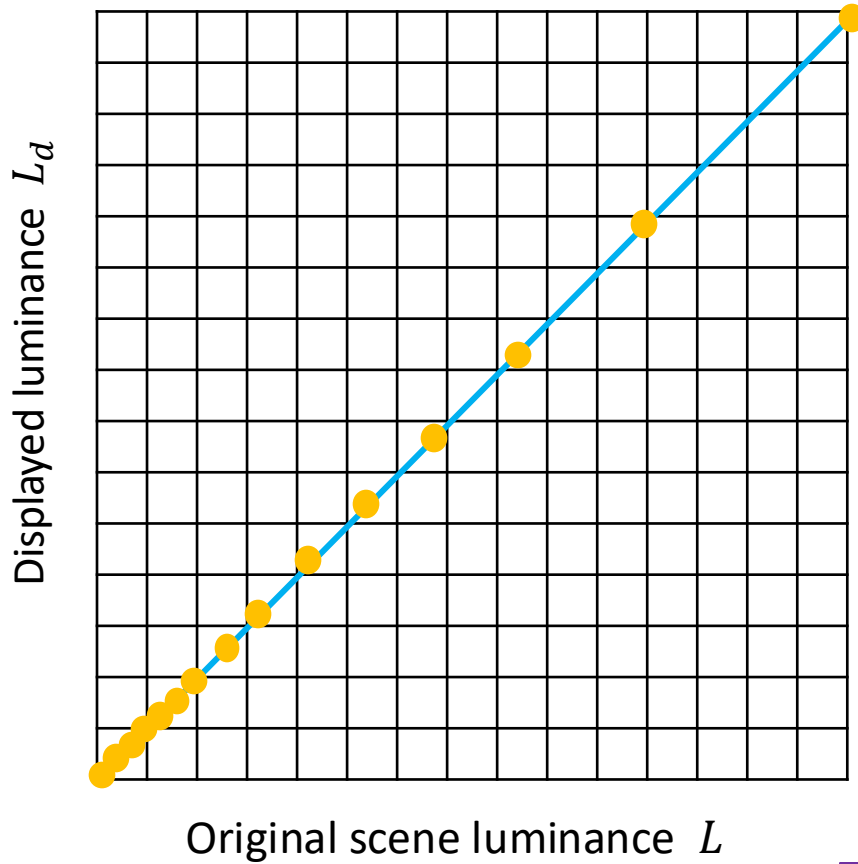


Original scene luminance L

3. End-to-end Effect

$$L_d = L$$

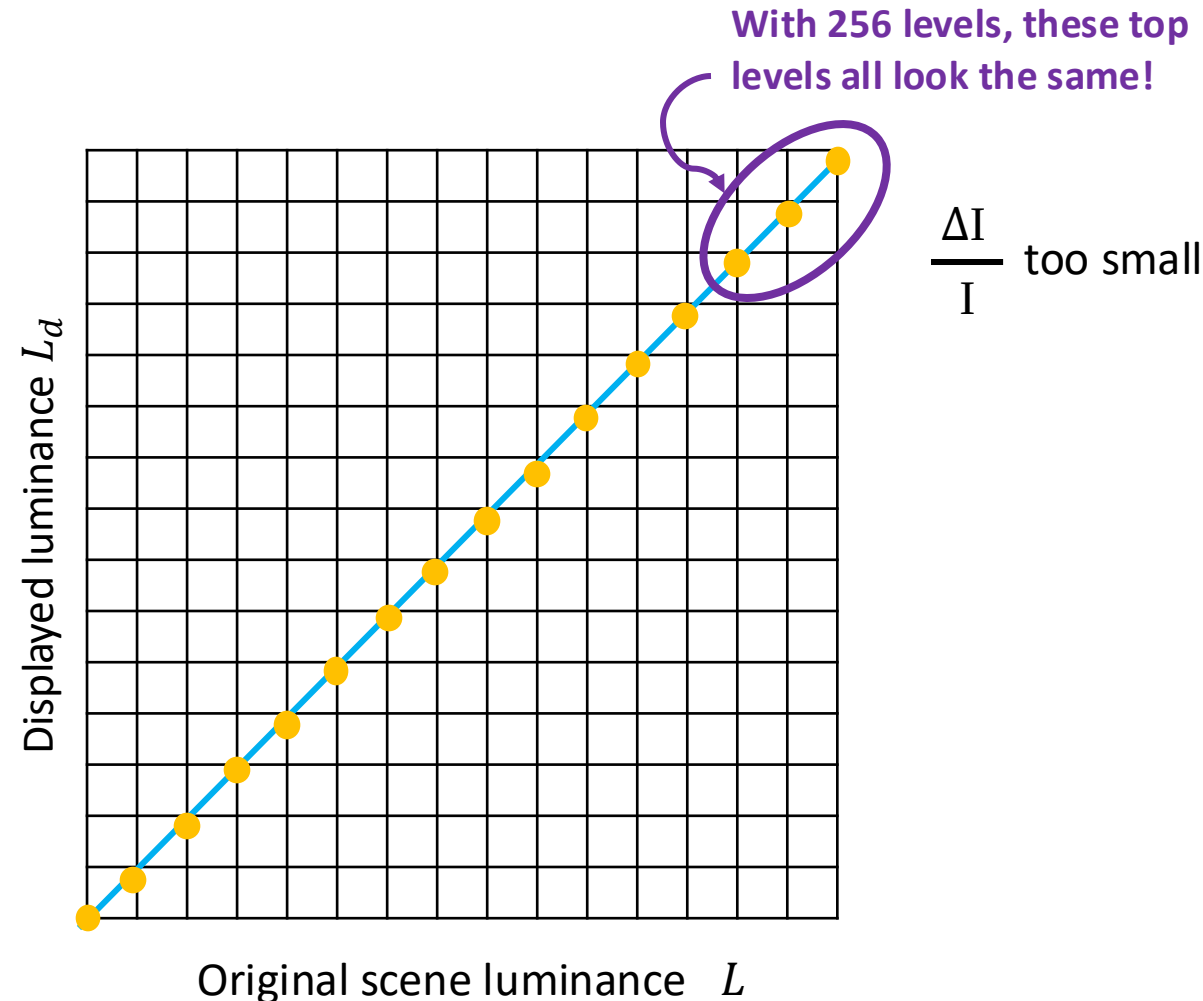
Gamma Encoding Comparison



Nonlinear (gamma) encoding

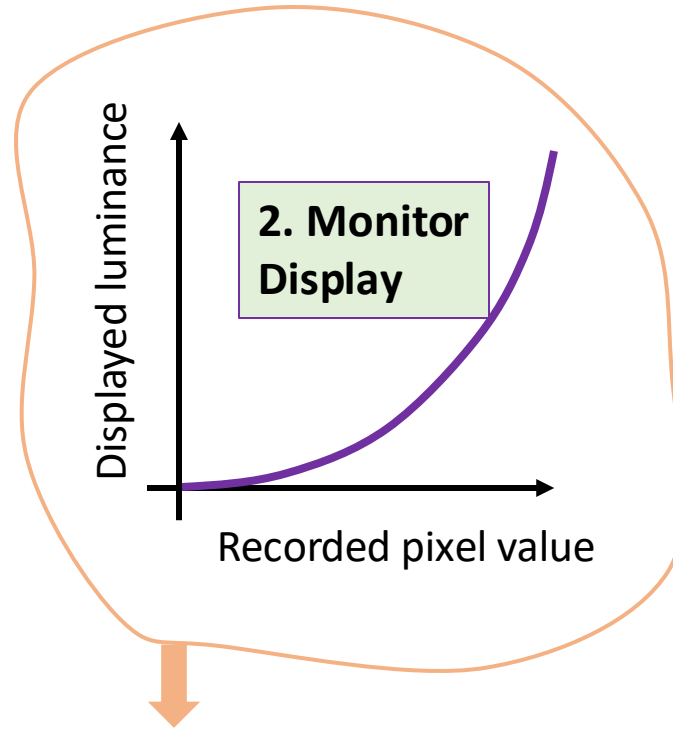
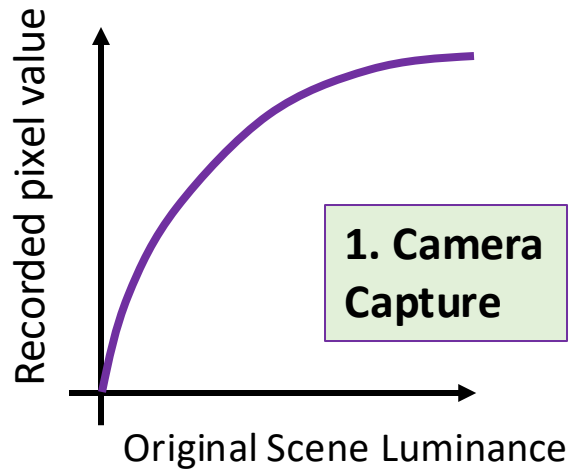
3. End-to-end Effect

$$L_d = L$$



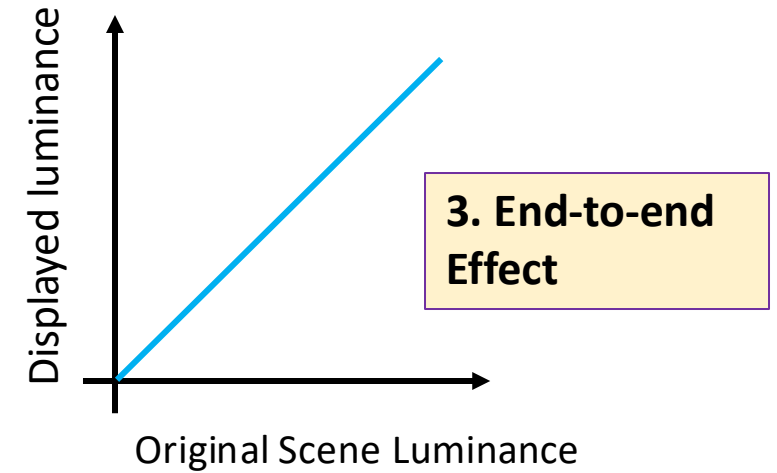
Linear encoding

Gamma Correction



$$L_d \propto v^\gamma$$

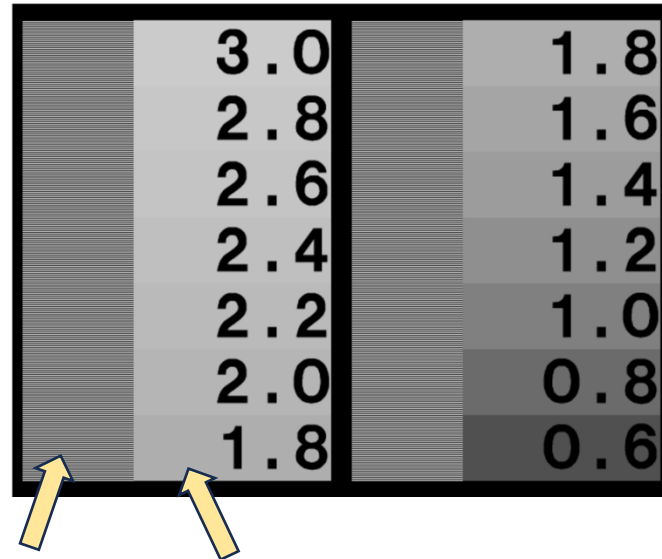
Can use a test image to see what the γ value is for your screen



Monitor must be running in its native resolution & image viewer must not resize the image.

Gamma Calibration Test Image

Not shown
correctly →
(resized in
powerpoint)



One section has alternating
black and white rows

Another section has
intermediate solid gray levels



If you stand back a bit, your eye will average
the black and white rows

$$L_d \propto v^\gamma$$

Ideally, if $\gamma = 1$, the black/white bars would
match gray level 128:

$$\frac{0 + 255}{2} \approx 128$$

while if $\gamma = 2$, the black/white bars would match
gray level 180:

$$\frac{0^2 + 255^2}{2} \approx 180^2$$

Last time

1. CIE RGB: (R_C, G_C, B_C) The monochromatic spectral primaries, used in the 1920's for the original color matching experiments
2. CIE XYZ: Used for color specification, all colors are in the positive quadrant
 - Reference system
3. For color display: RGB models
 - NTSC RGB
 - sRGB
4. Aiming at perceptual uniformity
 - UVW, $U^*V^*W^*$
 - Lab, $L^*a^*b^*$

Today

Systems for Color Image Processing

- Hue, Saturation, Intensity (HSI)
- Hue, Saturation, Value (HSV)
- Hue, Saturation, Lightness (HSL)

Gamma Correction

Systems for Color Compression & Transmission

- YIQ, YUV, YCrCb

Systems for Color Printing

- CMY, CMYK

Systems for Analog Color Transmission: YIQ, YUV

- Analog B&W TV, 1941, NTSC standard settled, first commercial broadcast in US
- 1946: CBS gave FCC first demo of their color TV
 - 72 frames/sec with R,G,B frame interleaving
 - Incompatible with B&W systems
- RCA flooded market with B&W sets to slow the potential launch of CBS color
- 1950: FCC approved CBS color TV system
 - It died
- 1953: FCC approved RCA color TV system
 - B&W compatible

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.211 & -0.523 & 0.312 \end{bmatrix} \begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix}$$

Y = luminance or luma

I,Q called inphase and quadrature components

- chromaticity information
- less important perceptually, could be sent in smaller bandwidth
- Why is it less important?

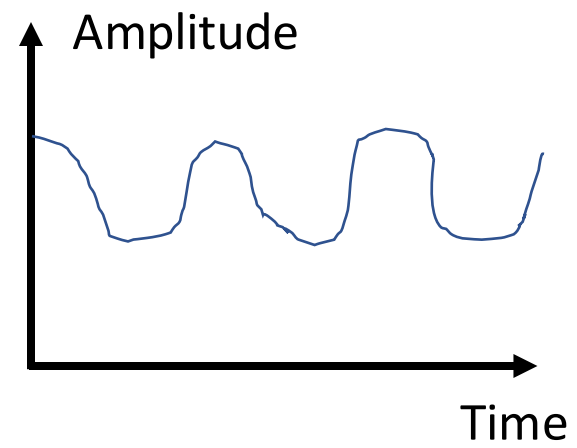
Notice: Sum of row 1 = 1

Sum of row 2 = 0

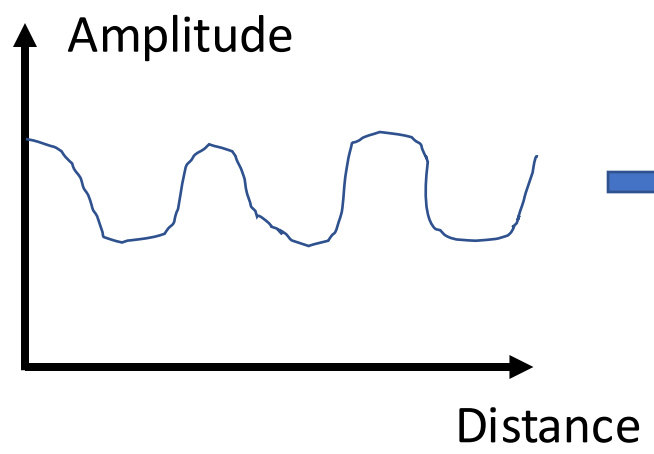
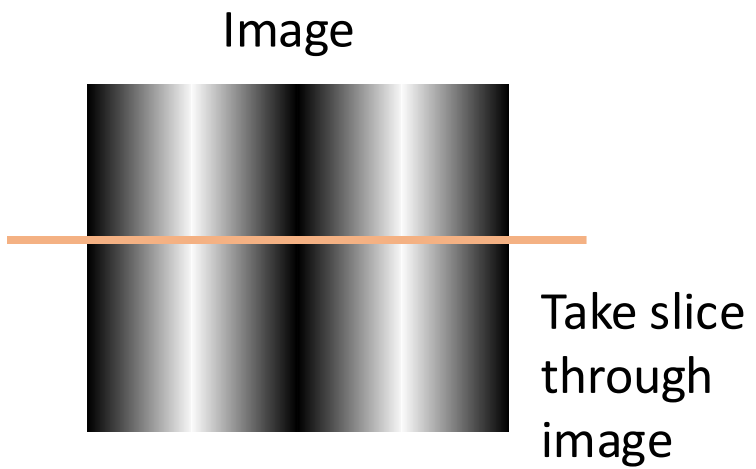
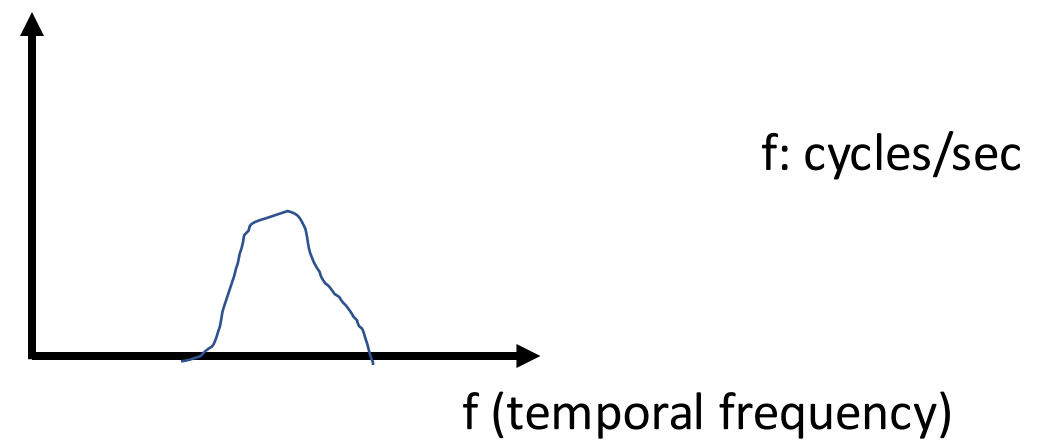
Sum of row 3 = 0

So: when R=G=B then I=Q=0, and Y=R=G=B

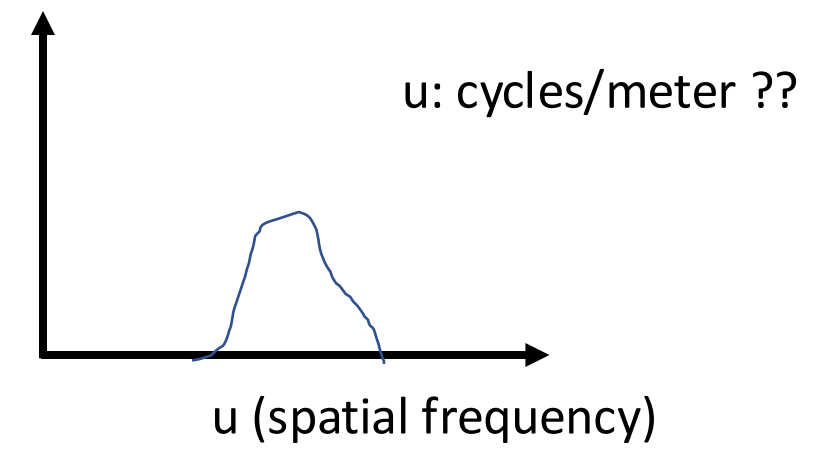
Temporal Frequency & Spatial Frequency



FT
→

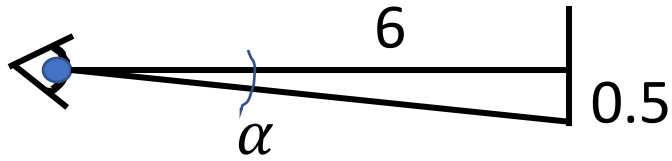


FT
→



Different Possible Units for Spatial Frequency

- Analogous to temporal frequency (cycles per second), spatial frequency can be measured in cycles per unit distance
- But we're also concerned how it looks to the eye:

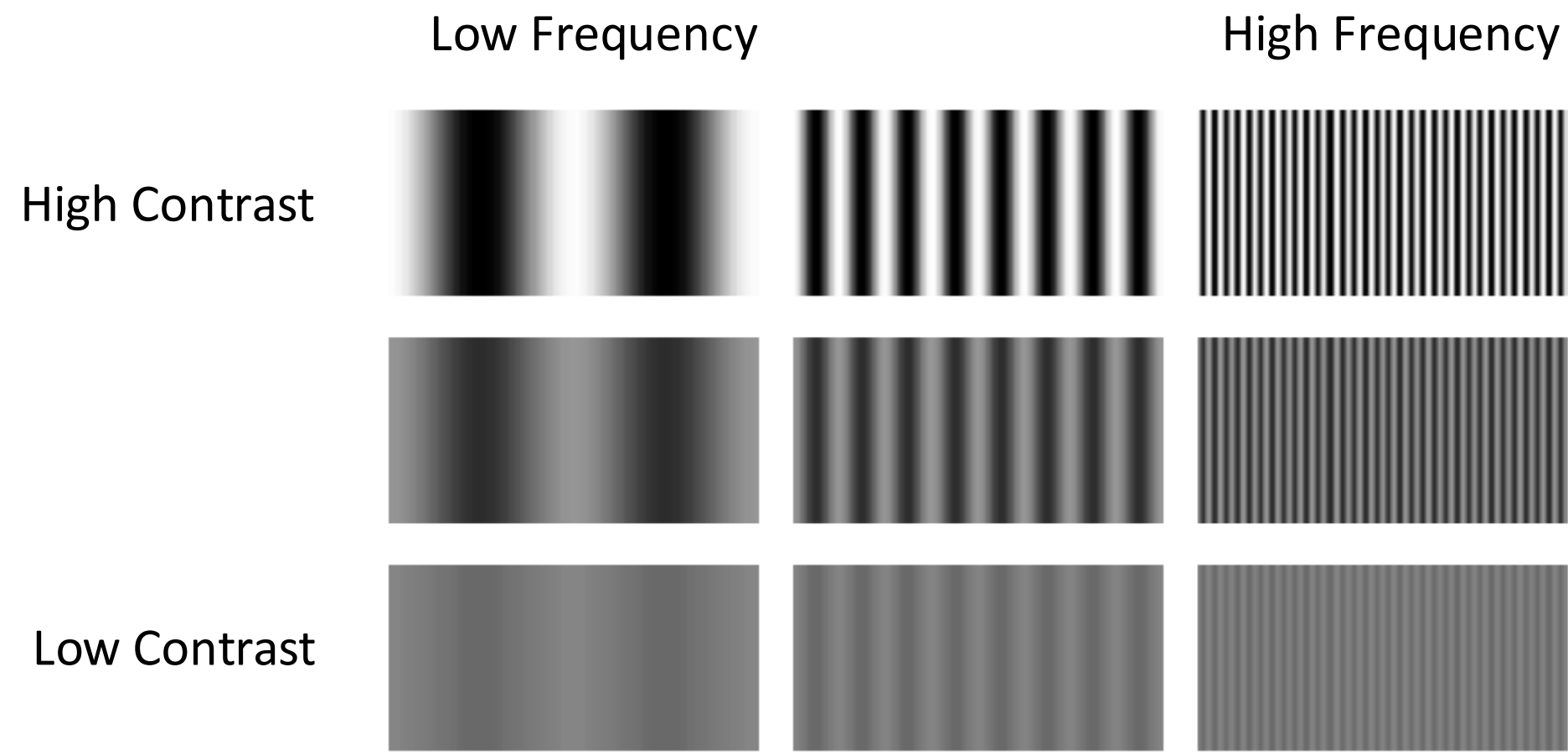


- $\tan \alpha = 0.5/6 = 1/12$
- $\alpha \approx 4.76$ degrees

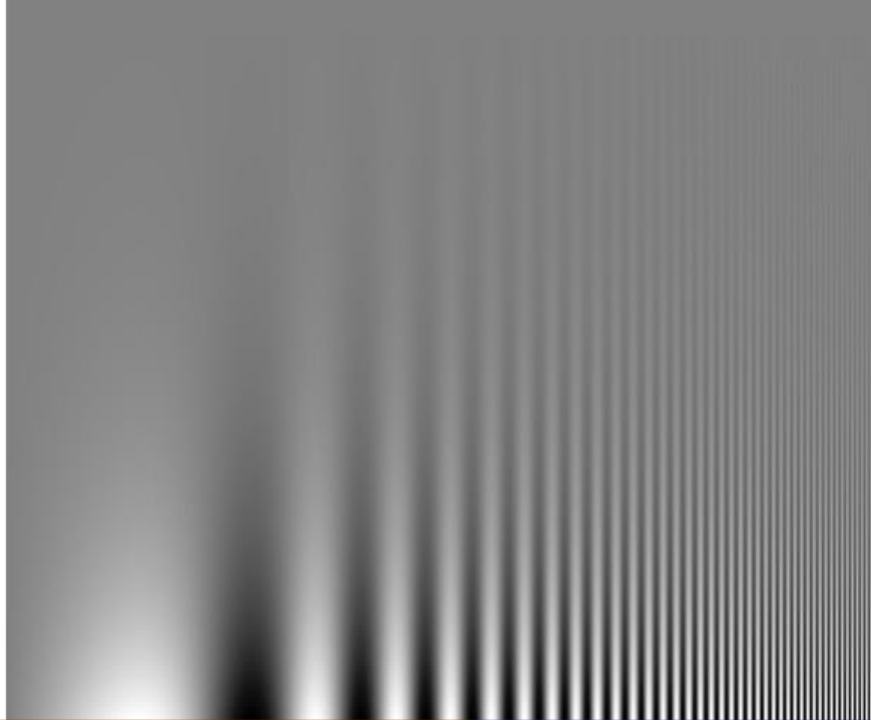
- Suppose half screen width = 3600 pixels
- Suppose some pattern has 1 cycle per 320 pixels

$$\frac{3600 \text{ pixels}}{4.76 \text{ degrees}} \times \frac{1 \text{ cycle}}{320 \text{ pixels}} = 2.36 \text{ cycles per degree of visual angle} = 2.36 \text{ cpd}$$

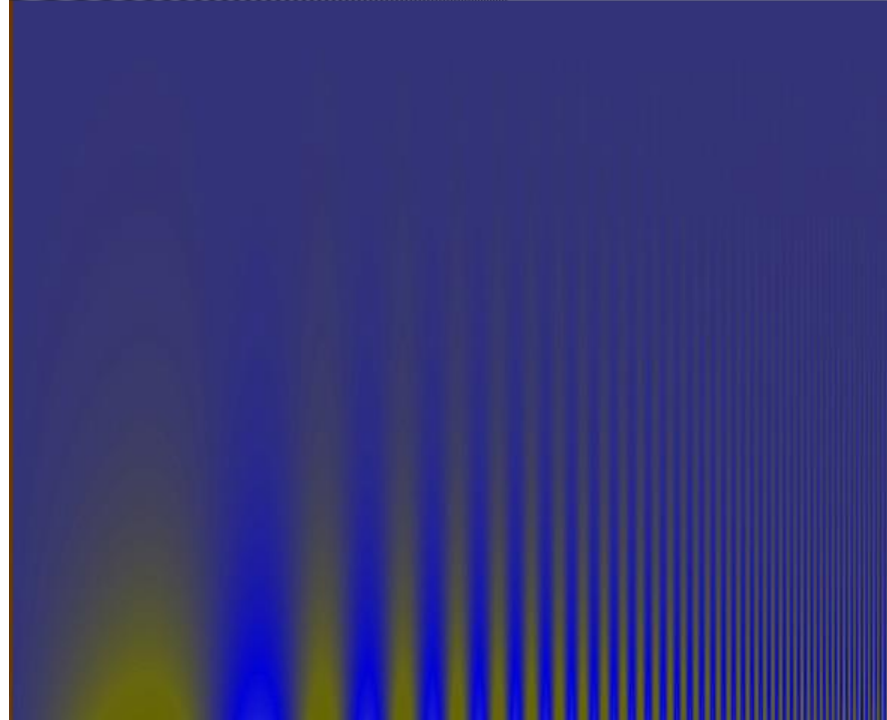
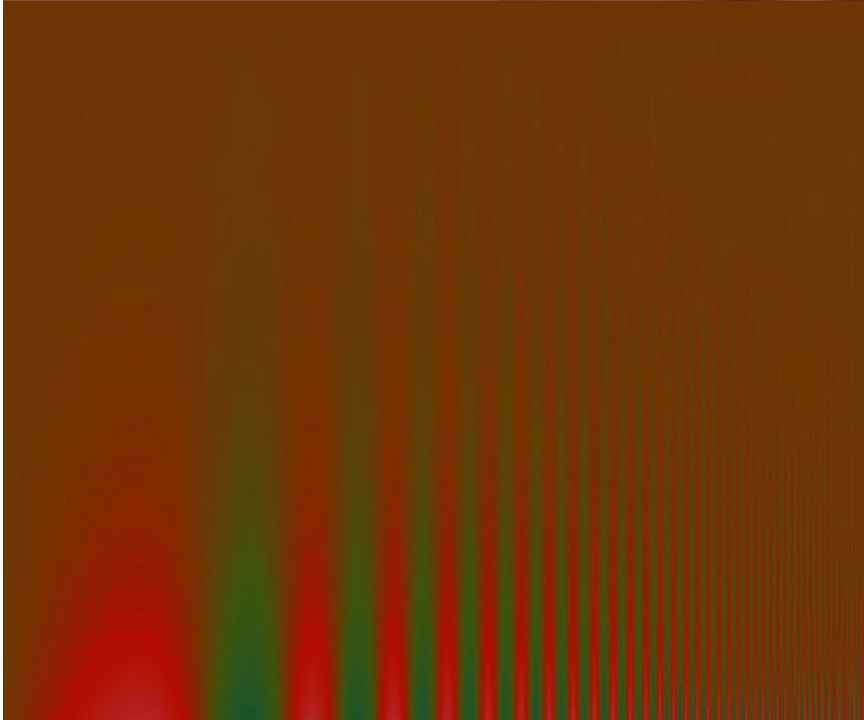
Sensitivity to Spatial Frequency



Spatial Frequency Sensitivity



- Human visual system is not sensitive to high spatial frequencies
- And is even less sensitive to high spatial frequencies when they are in color



Color Systems for Video Transmission: YIQ & YUV (analog), YCbCr (digital)

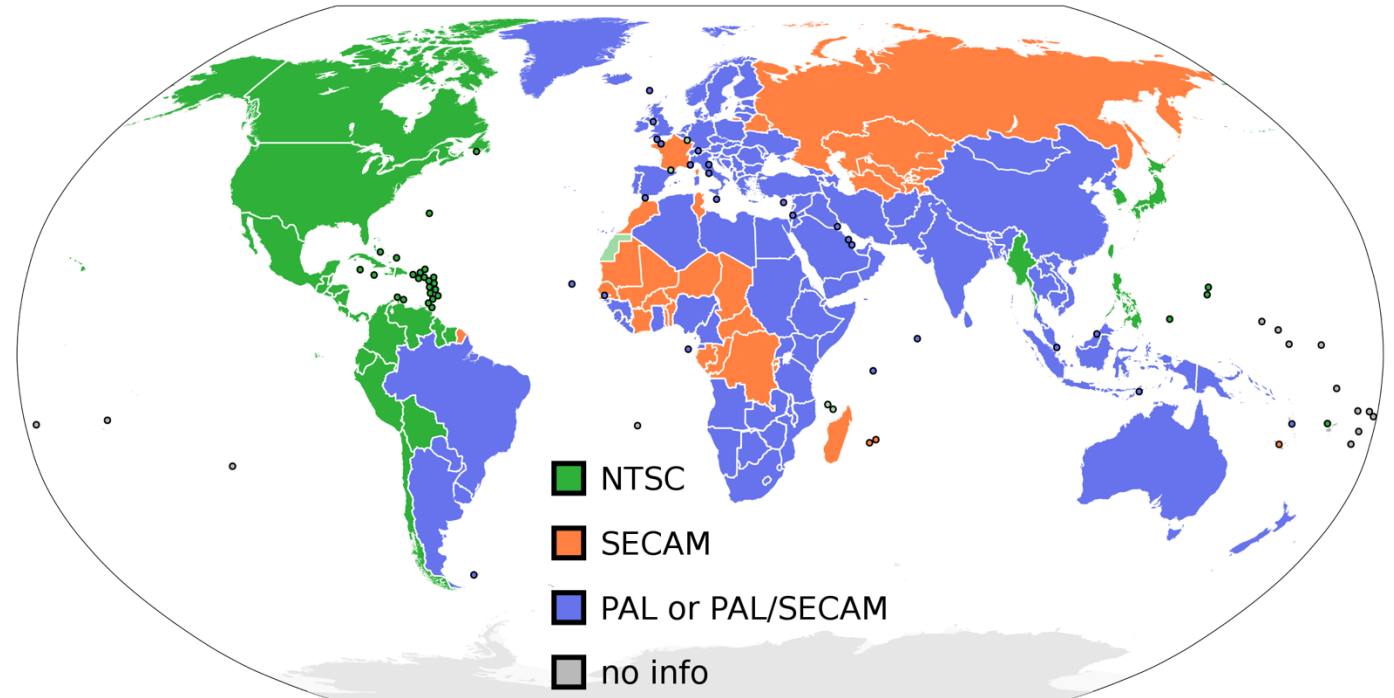
- Elsewhere: PAL, SECAM → YUV
- *Sequential Color with Memory*

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = [3 \times 3] \begin{bmatrix} R_N \\ G_N \\ B_N \end{bmatrix}$$

- SECAM = “*Système Essentiellement Contre les Américains*”

NTSC = Never Twice the Same Color

- Digital storage application: JPEG & MPEG Compression
- YCbCr: similar idea, chromaticity planes get downsampled



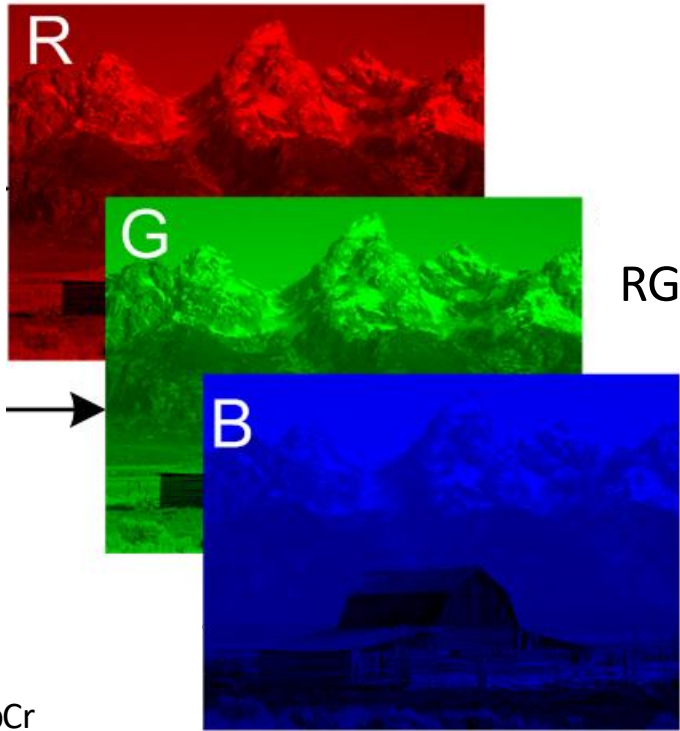
These systems:

- have one color plane for luminance/intensity
- have two color planes for chromaticity information
- chromaticity given less bandwidth or fewer bits

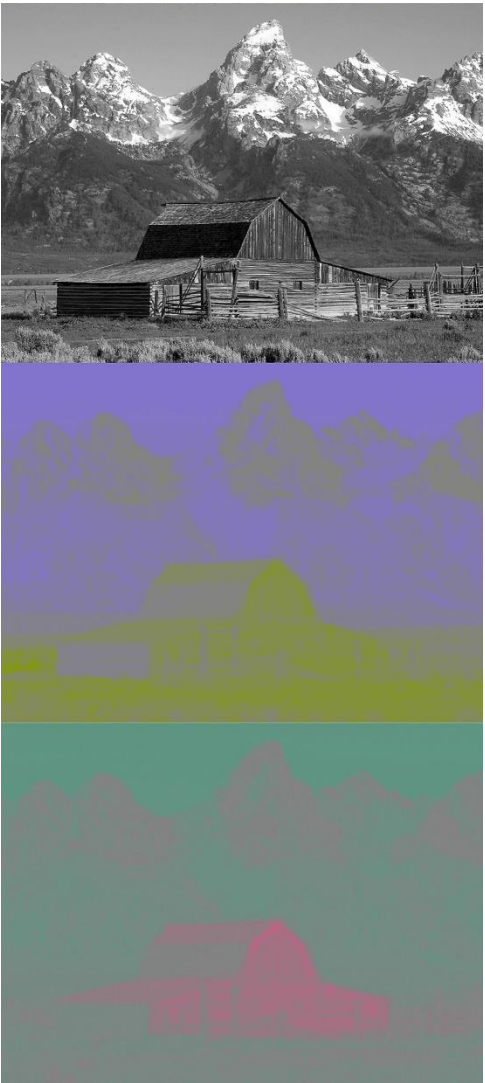
How does this color system help?



Pictures from:
<https://en.wikipedia.org/wiki/YCbCr>



RGB → YCrCb

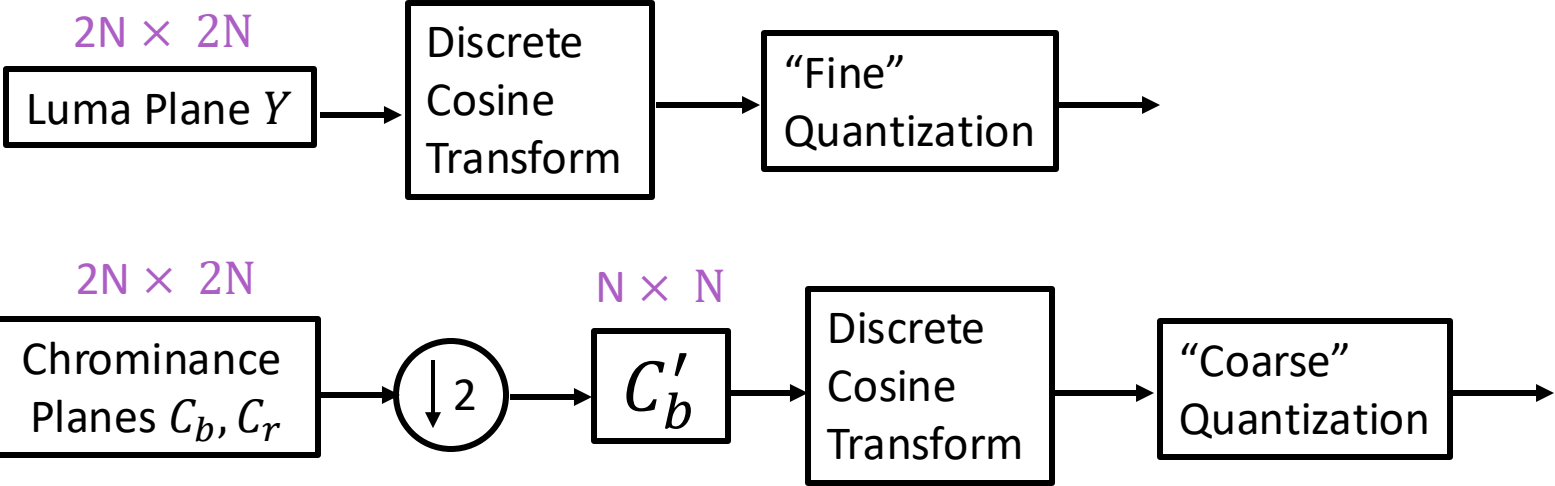


Luma
Component
Y

Cb

Chroma
Components

Cr



Subtractive color matching

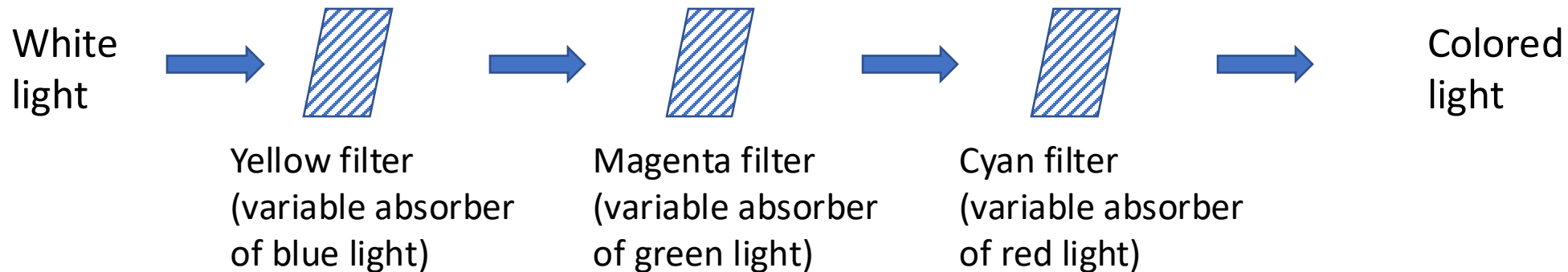
1. So far: Additive color matching:

- Start with zero intensity of 3 primaries = Black
- Progressively add more light → Eventually get White

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

2. Subtractive color matching:

- Start with white light, nothing filtered out
- Progressively filter out more → Eventually get Black



Color Printing Systems: CMY & CMYK

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

In practice, need to have a separate black ink (K)

- Reduces need for large amounts of colored ink (cost)
- Impurities in the ink
- Thickness of build-up
- Reflections off surface of ink (instead of passing through all layers)

Converting from CMY to CMYK:

$$K = \text{Black Ink}$$

When R=G=B=0 in the RGB system, this corresponds to pure black:

- C=M=Y=1 in the CMY system
- K=1 and $C_{CMYK} = M_{CMYK} = Y_{CMYK} = 0$ in the CMYK system

Otherwise:

$$K = \min(C, M, Y) \quad C_{CMYK} = \frac{C - K}{1 - K}$$

$$M_{CMYK} = \frac{M - K}{1 - K} \quad Y_{CMYK} = \frac{Y - K}{1 - K}$$

Summary of color coordinate systems

1. CIE RGB: (R_C, G_C, B_C) Monochromatic spectral primaries, used for the original color matching experiments
 2. CIE XYZ: Used for color specification, all colors are in the positive quadrant
 3. Display-oriented models: RGB
 - NTSC RGB
 - sRGB
 4. Color spaces that aim at perceptual uniformity
 - UVW, $U^*V^*W^*$
 - Lab, $L^*a^*b^*$
 5. Models for color printing
 - CMY, CMYK
 6. Models for efficient transmission and storage
 - YIQ, YCbCr, YUV
 7. Perceptually-based models
 - HSI, HSV, HSL
-   represent ways of capturing intensity / brightness / luminance / lightness
- Note: The Y in XYZ is the same Y as in YIQ, YUV, YCbCr, and same as V in UVW (all marked )

They are not the same Y as in CMY & CMYK